

EasyVisa - Analysis and Solution

EasyVisa Project for AI & ML

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Executive Summary - Insights

- We can see that a lot of the applicants with high education levels are getting accepted at a high rate. Having only a high school education is a major reason why most applicants are getting denied
- We see that the Midwest and South regions approve the most applications
- Europe, Africa and Asia have the most certified applications
- Prevailing wage is lower in importance than Education level

All these insights are very important when looking at if someone gets accepted or denied for a visa.

Executive Summary - Recommendations

- Create a pre-check process. When people apply and select high school as their highest education level, the system should give them a heads up that their case is unlikely to be certified. This will reduce the number of applicants as a whole and save resources that would be used reviewing these applications
- Applicants from Asia applying for the South region are 2 important factors for the model. If an applicant fits this criteria have them go to an automatic review based on their expected wage. For example if someone with a Doctorate degree from Asia applies to the south region but the job offers a wage lower than the regional median then this will be flagged for review by a person. This process will protect certified applicants and the integrity of the labor market. If the wage is a fair wage they will be accepted without review
- Because the model has a 93.7% recall, the company should implement a system where if an applicant receives a certified prediction with a high confidence score over 90% they should be move to a fast track automated pipeline. This will reduce the work load for the company and they will then be able to focus on applications that do not fit this criteria

Business Problem Review and Solution Approach

- The Office of Foreign Labor Certification (OFLC) processes certification applications for employers to bring foreign workers to the United States
- The number of applicants is increasing every year making this a tedious task for OFLC and they need a machine learning based solution streamlining the process
- We need to facilitate the process of visa approvals and create a model that can accept or deny applicants using the criteria OFLC follows

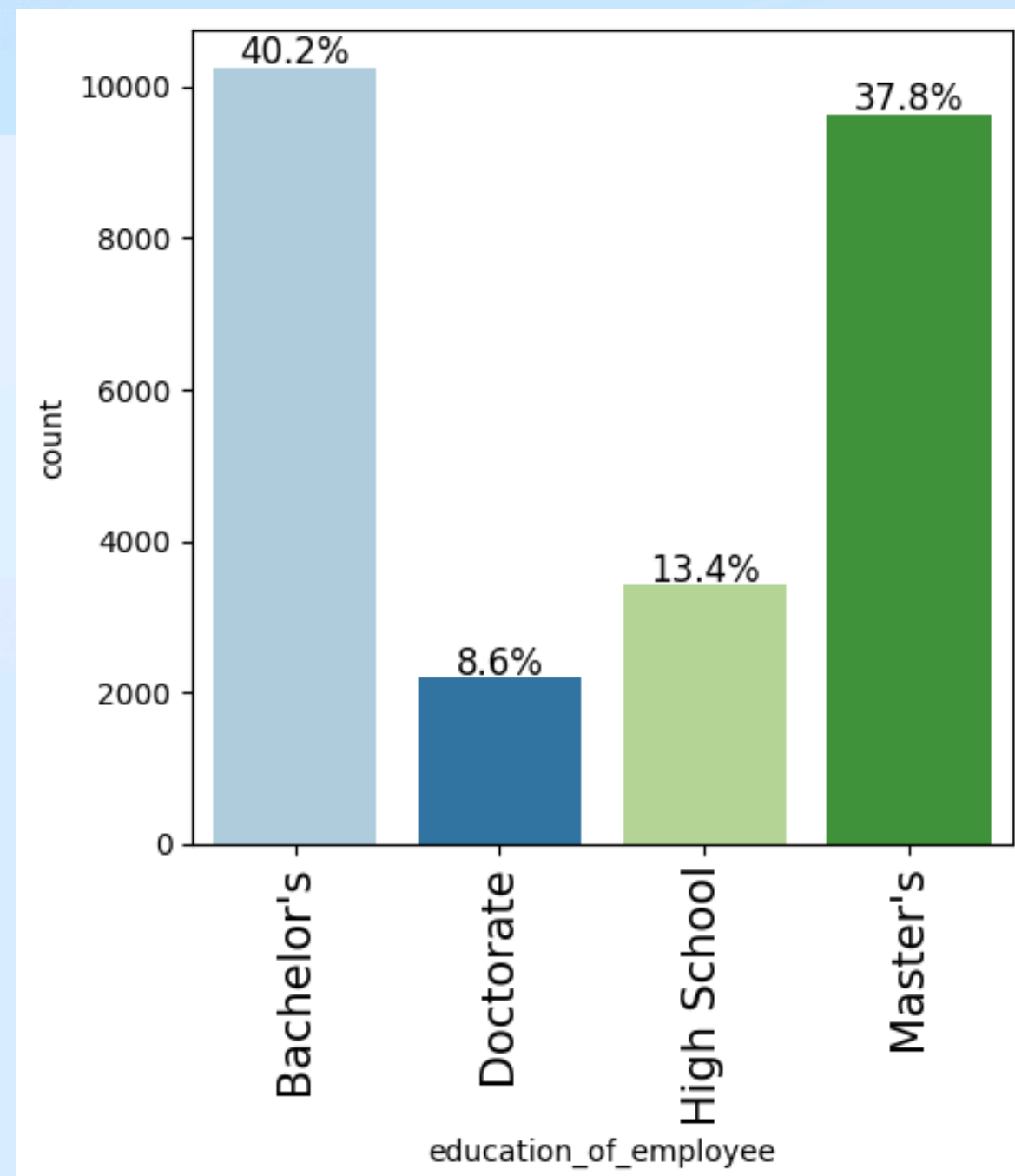
EDA Results

- We see there are 12 columns and 25,480 rows in the dataset
- The majority of applicants come from Asia which is 16,861 applicants. Oceania has the least number with 192 applicants
- Majority of applicants have at least a bachelor's degree
- 10,678 of applicants have 0 job experience
- 17,018 of applications were certified

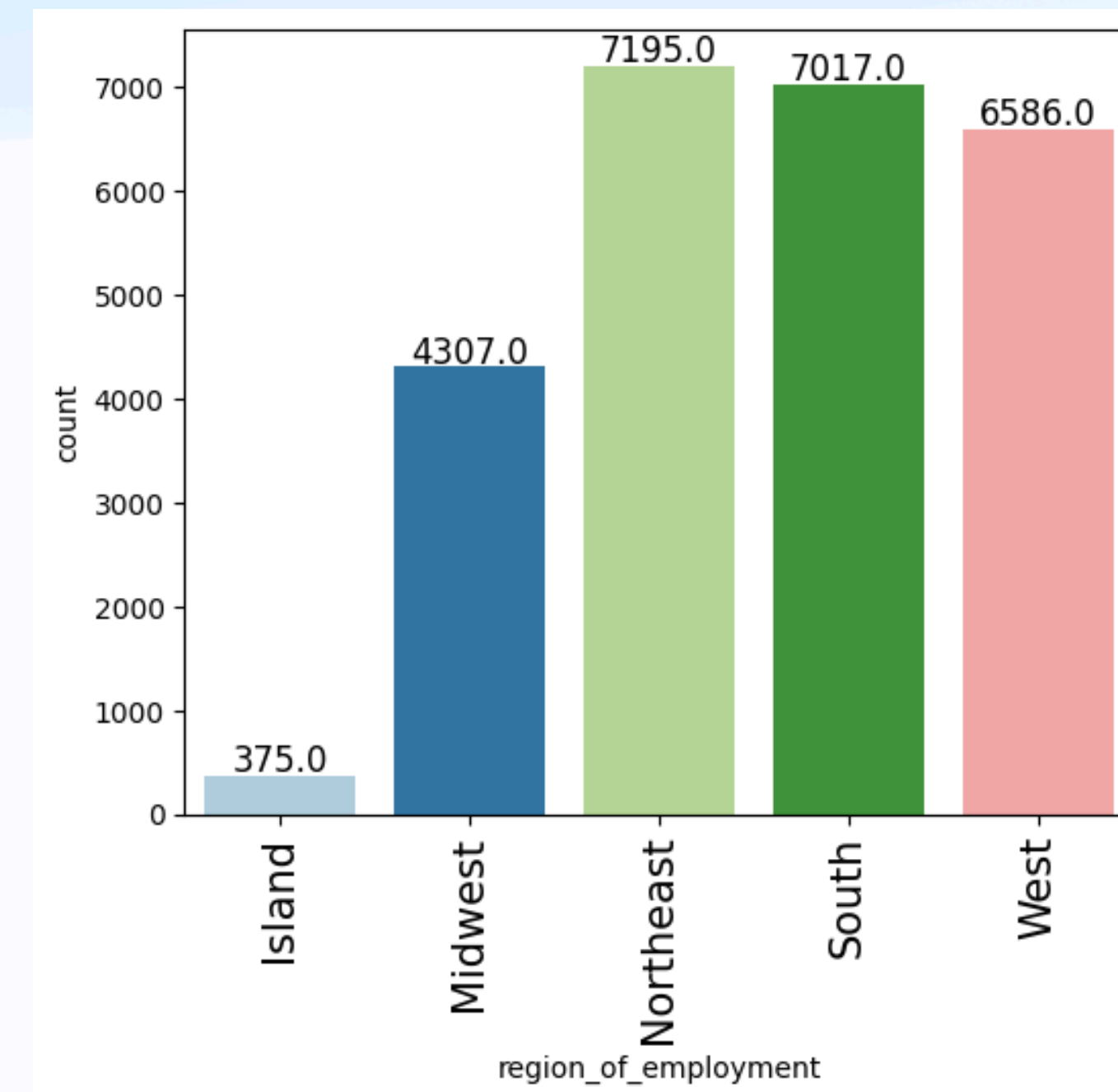
EDA Results - Univariate Analysis

Education of Employee & Region of Employment

We see the about 22,000 employees have at minimum a Bachelor's degree.



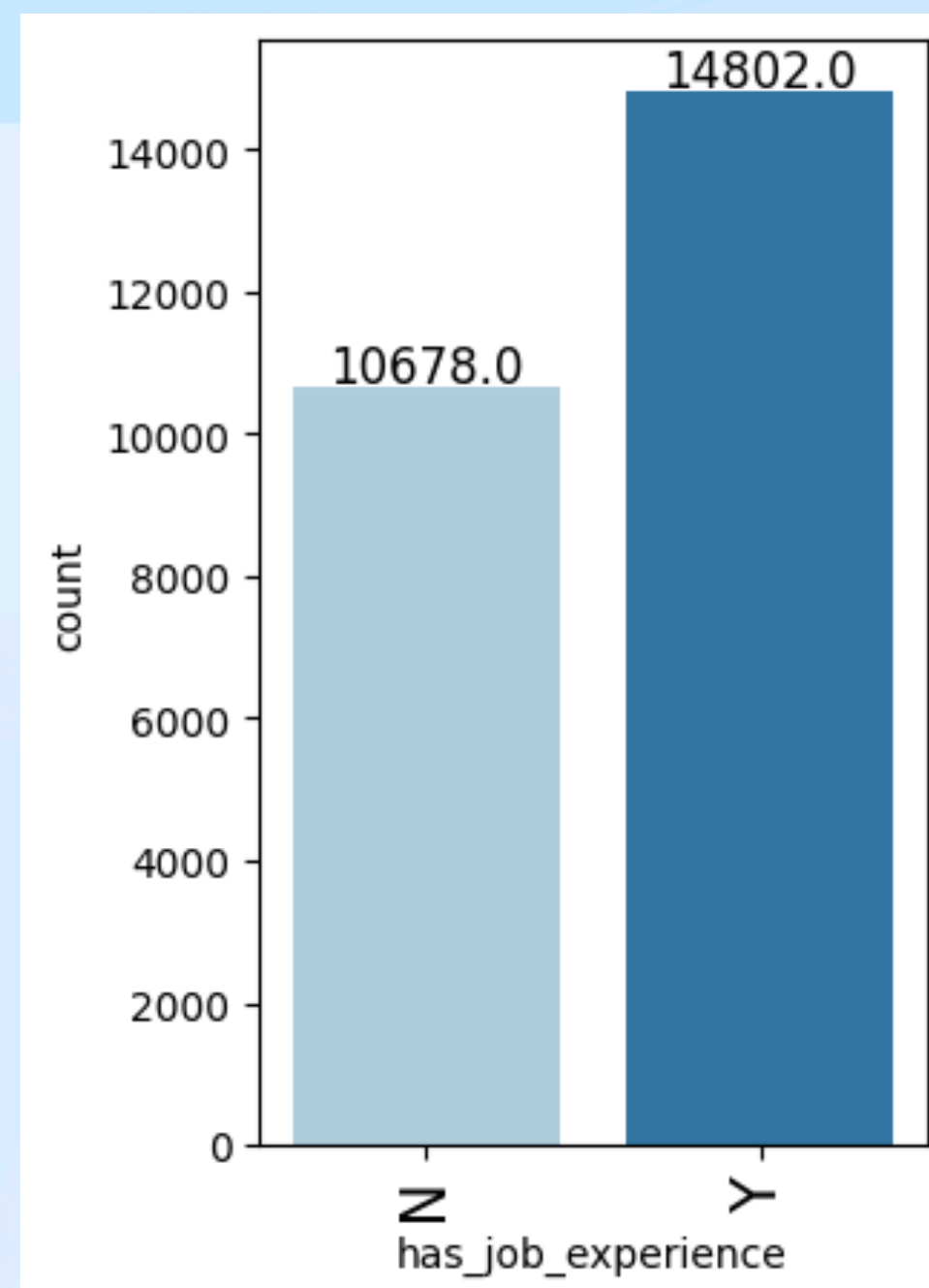
We see the applicants are applying throughout all of the USA with the top 2 regions being the Northeast and South USA.



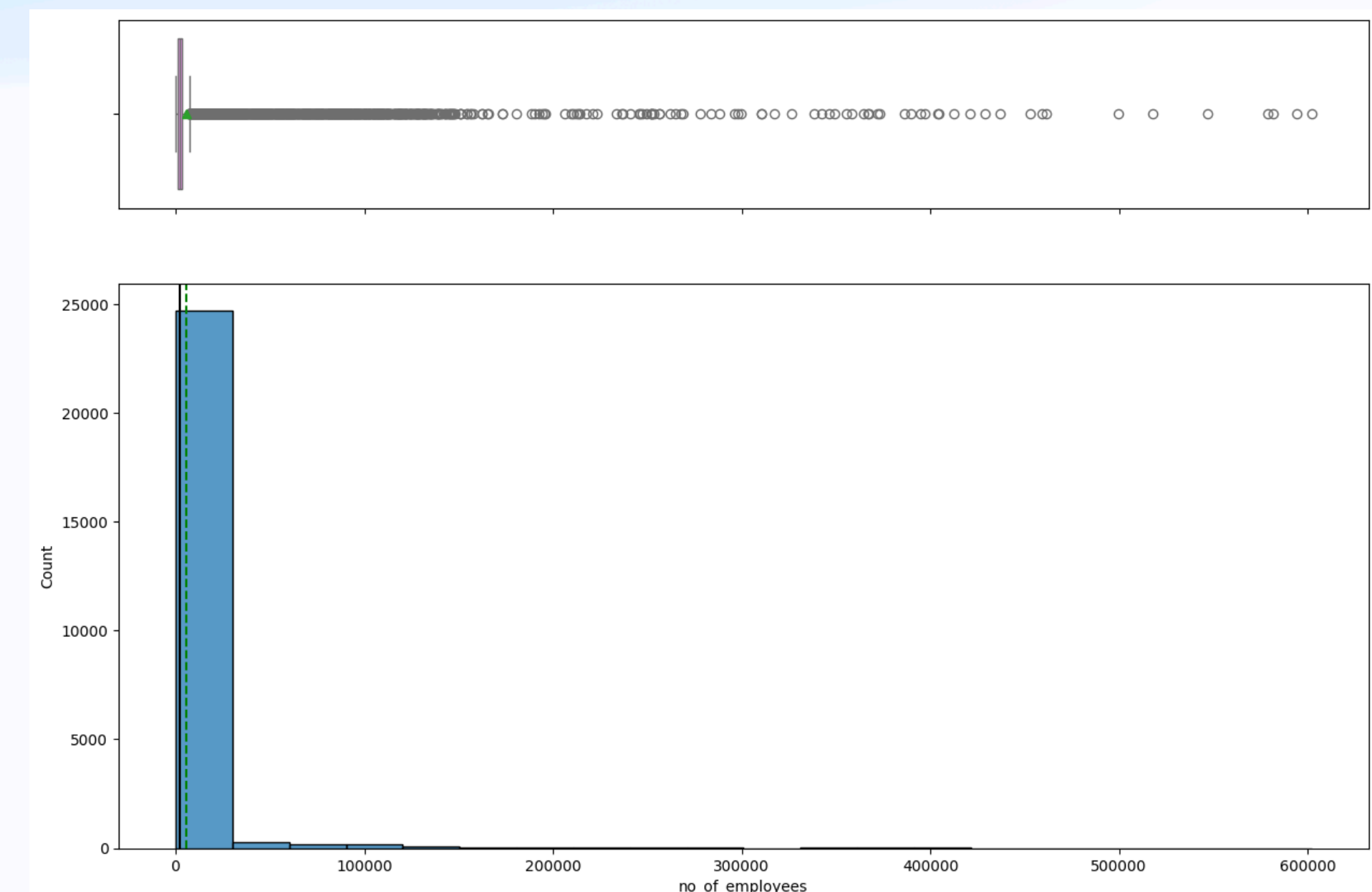
EDA Results - Univariate Analysis Cont.

Job Experience & Number of Employees

We see the majority of applicants have job experience. A large number of them do not have job experience.



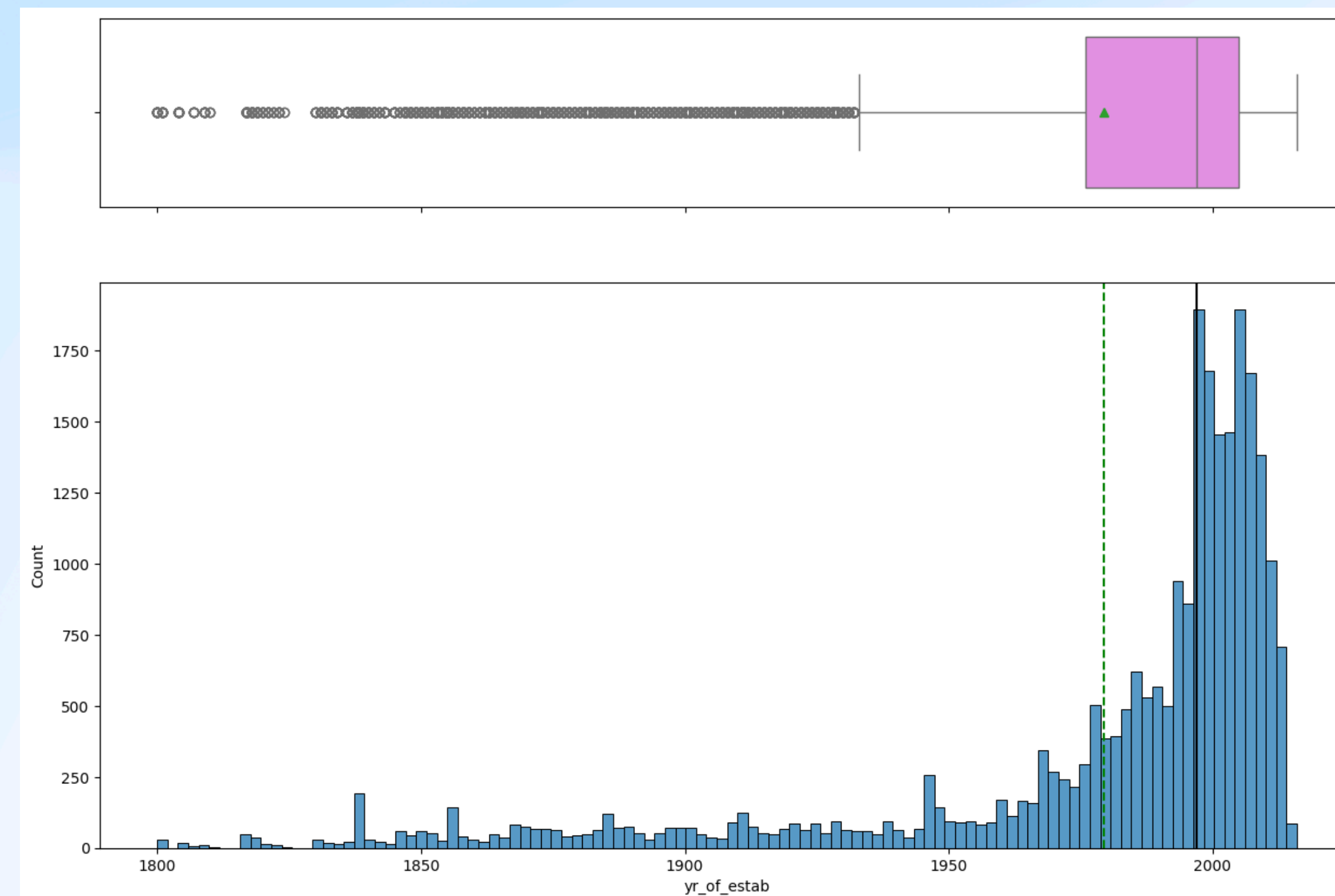
We see 75% of the total number of employers have 3,500 or less employees. The max number of employees one employer has is ~602,000 employees.



EDA Results - Univariate Analysis Cont.

Year of Establishment

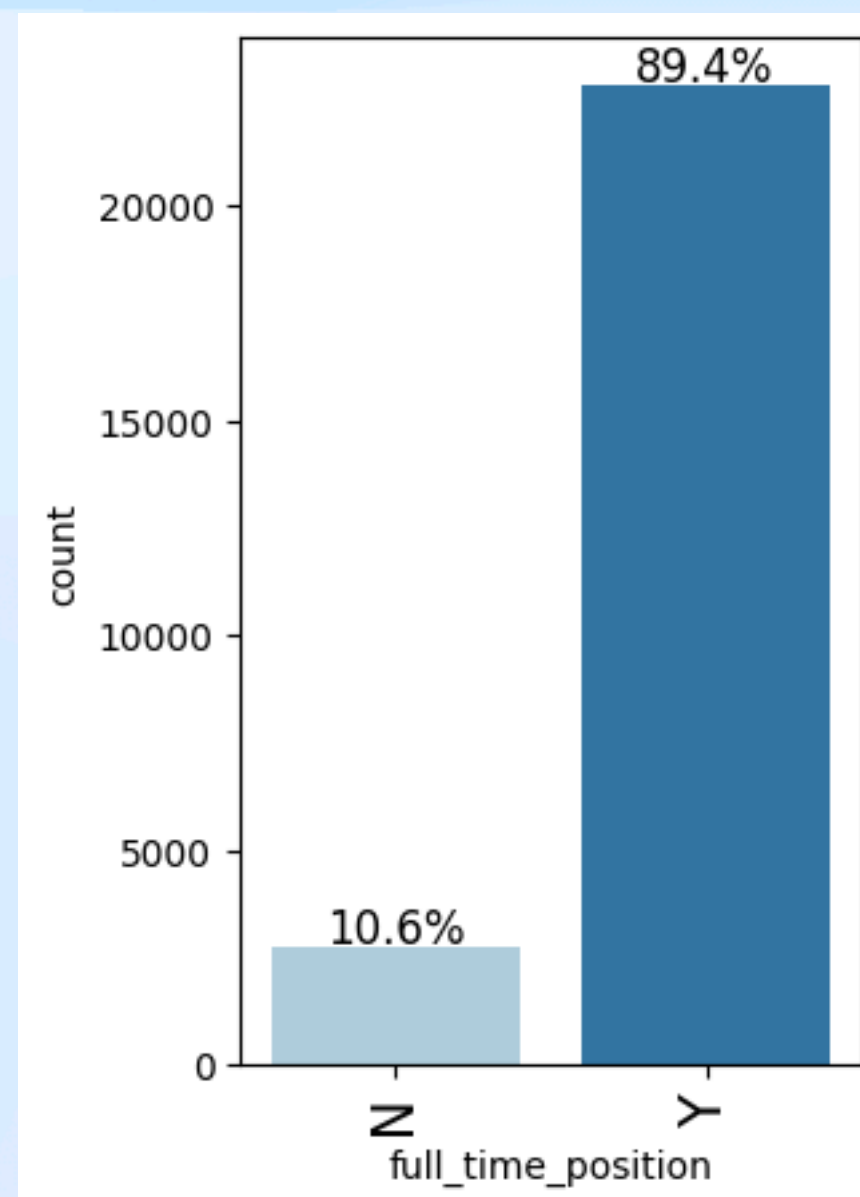
We see the graph is left skewed. The oldest company was founded in 1800 and the youngest company was founded in 2016. 75% of all companies were founded on or before 2005. The average year a company was founded was in 1979.



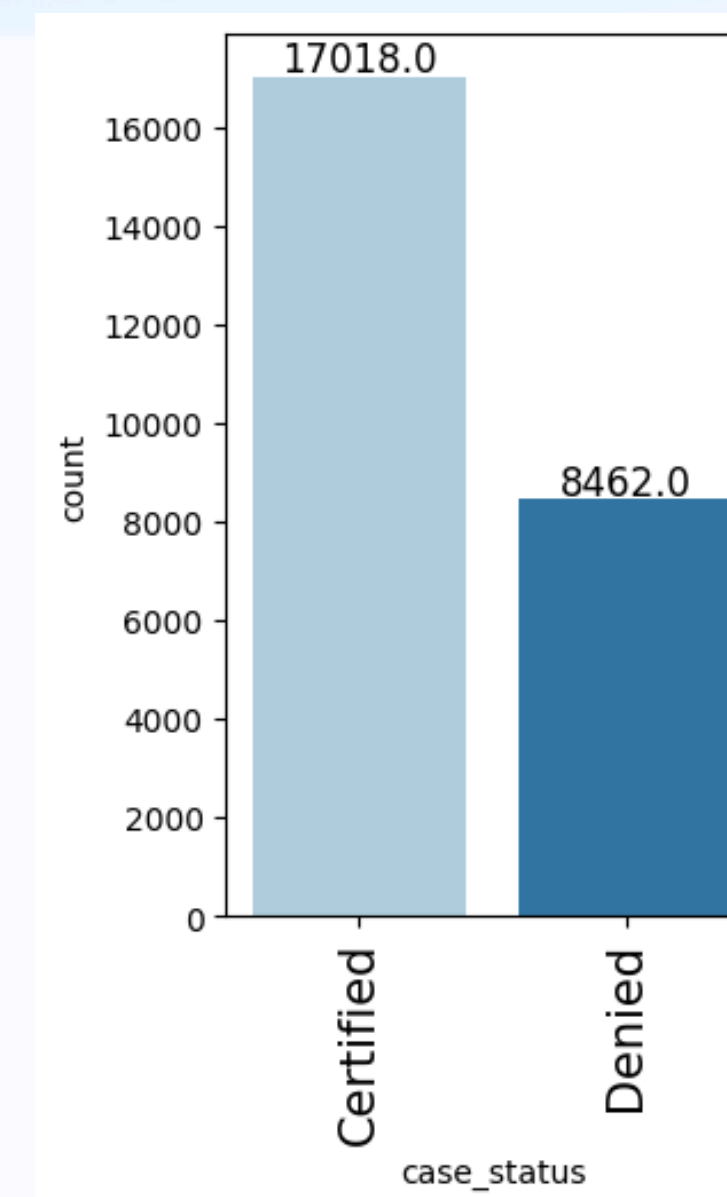
EDA Results - Univariate Analysis Cont.

Full Time Position & Case Status

We can see that over 20,000 applicants are seeking full time employment. This is 89.4% of the data.

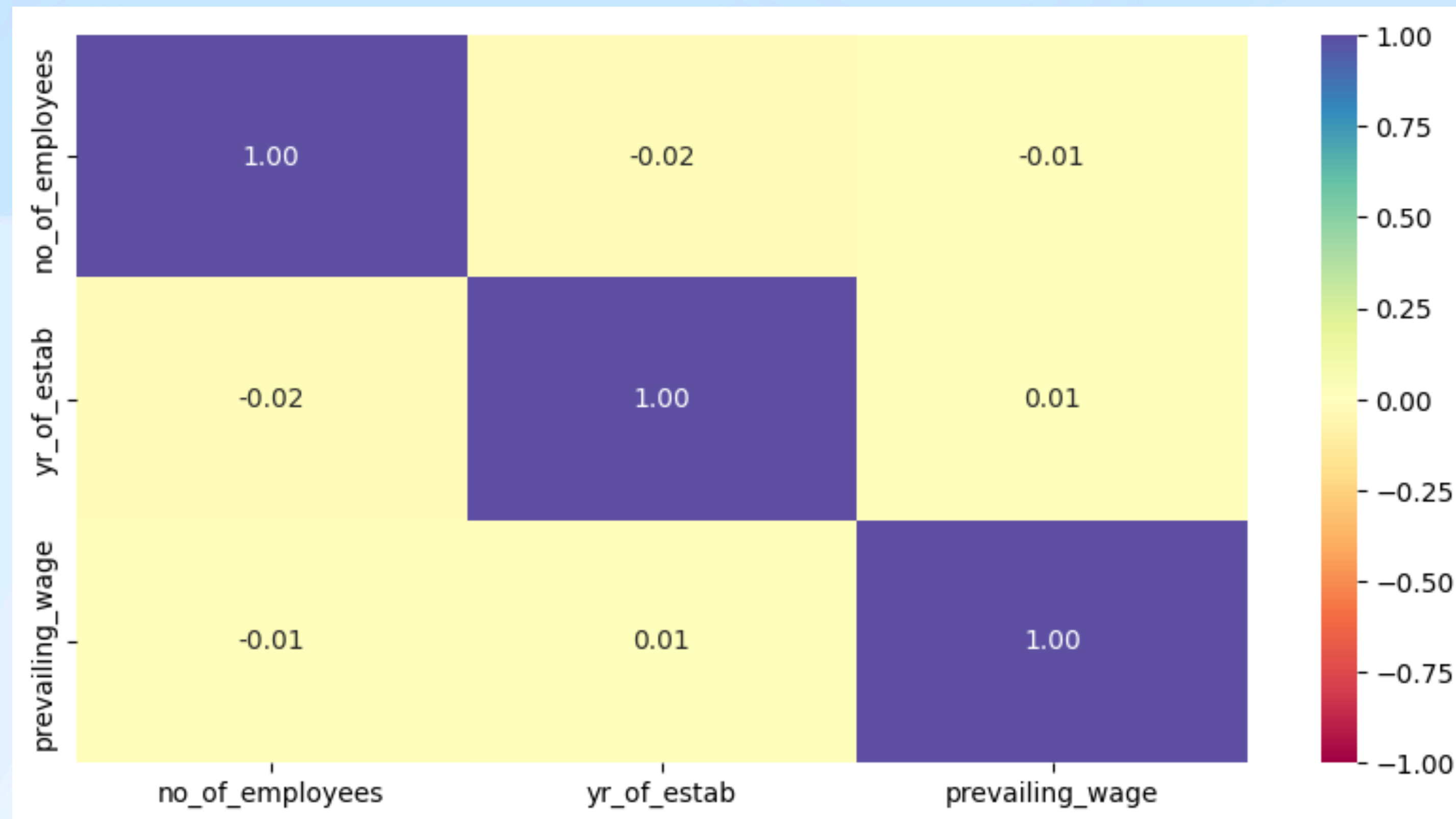


We see that 17,018 of all visa applications were certified and 8,462 were denied. 66.8% were certified to be exact.



EDA Results - Bivariate Analysis

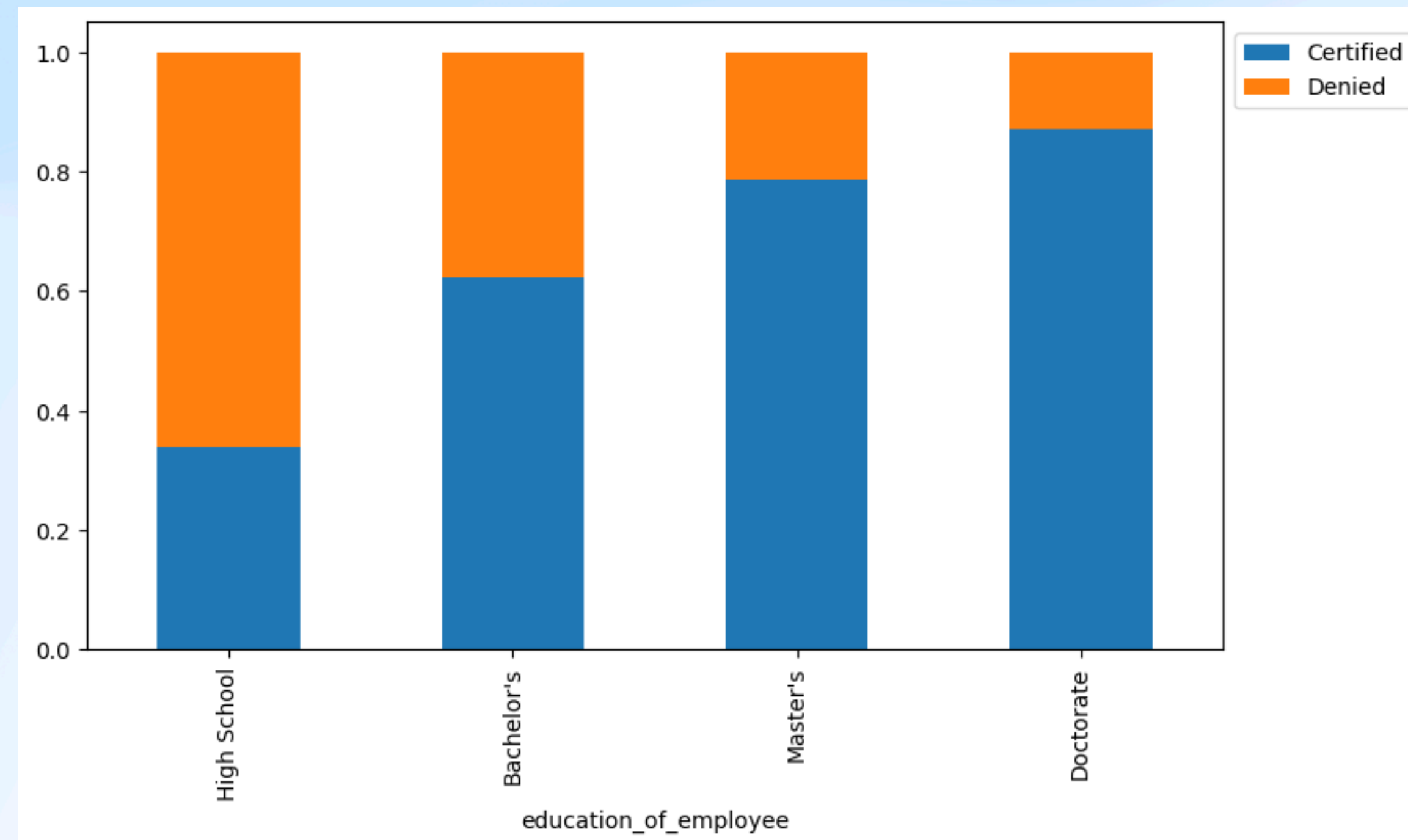
Looking at some of the columns we see that there is not any correlation between them.



EDA Results - Bivariate Analysis

Education Levels vs Case Status

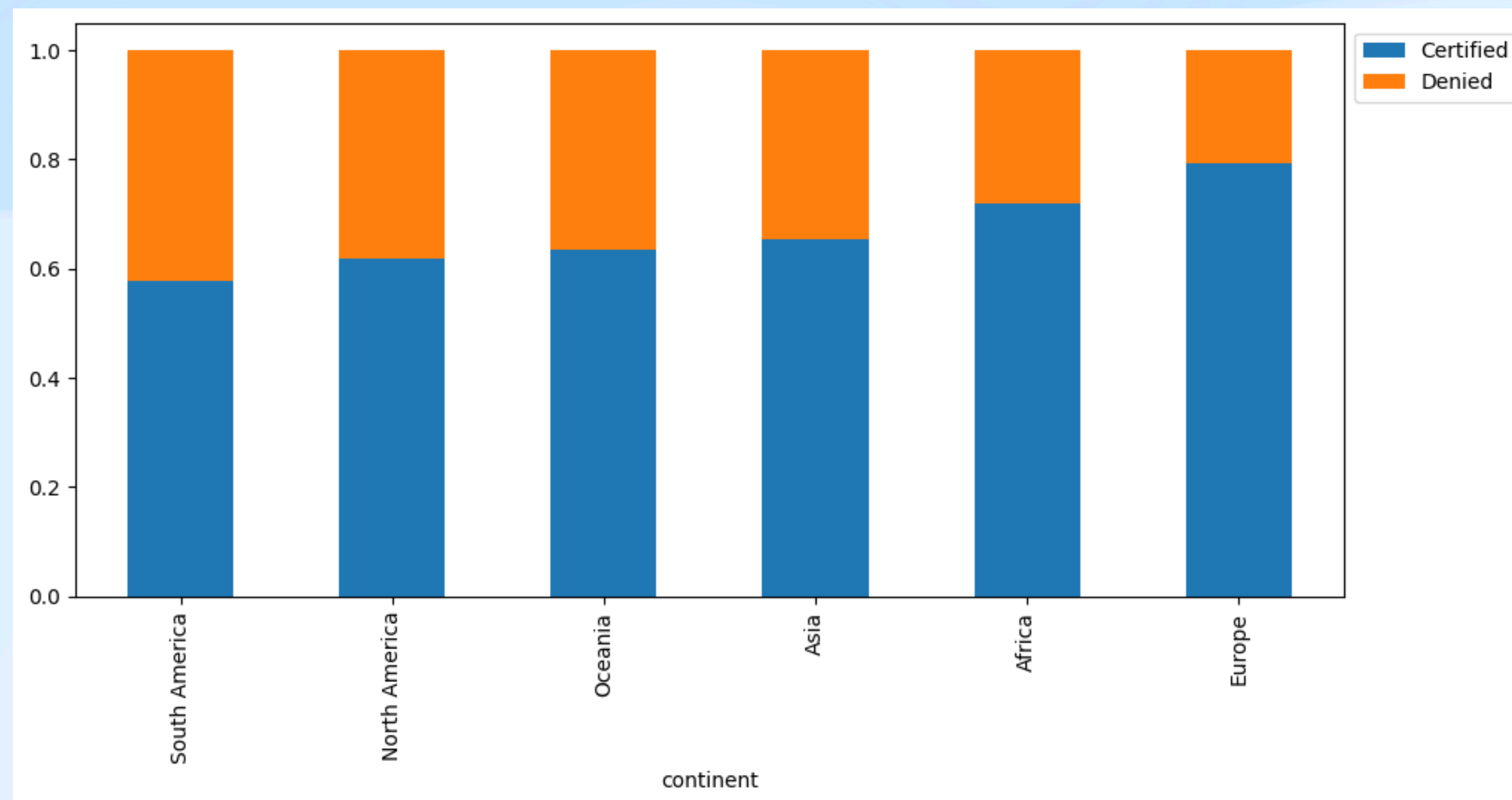
We see that the higher the education level the more likely they get approved for a visa. Doctorate education level applicants are at about a 90% approval rate.



EDA Results - Bivariate Analysis

Continent of Origin vs Case Status

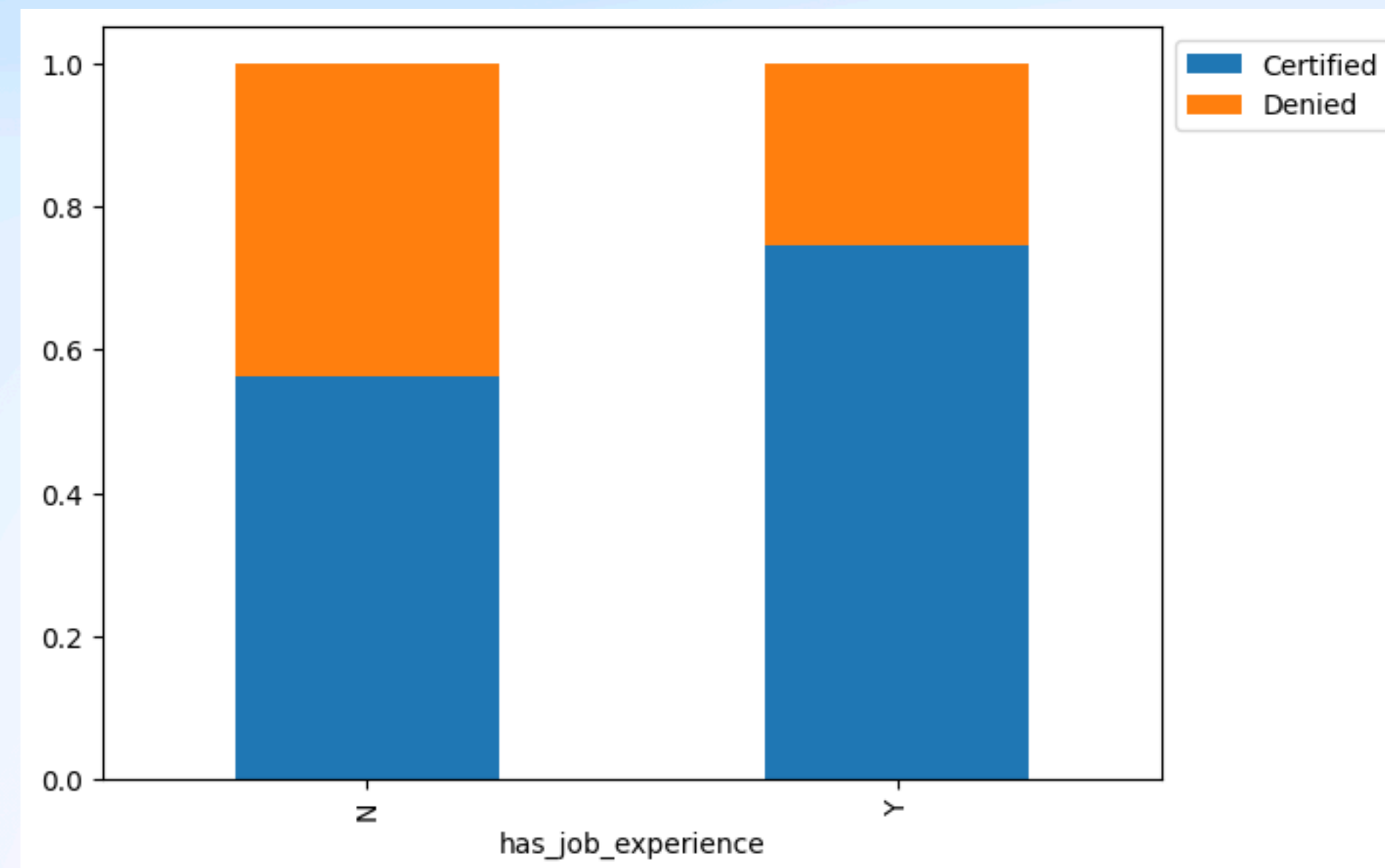
Visa acceptance rates do vary by continent with Europe and Africa leading the way. South America has the lowest approval rate with ~58% and Europe has the highest with ~75%.



EDA Results - Bivariate Analysis

Job Experience vs Case Status

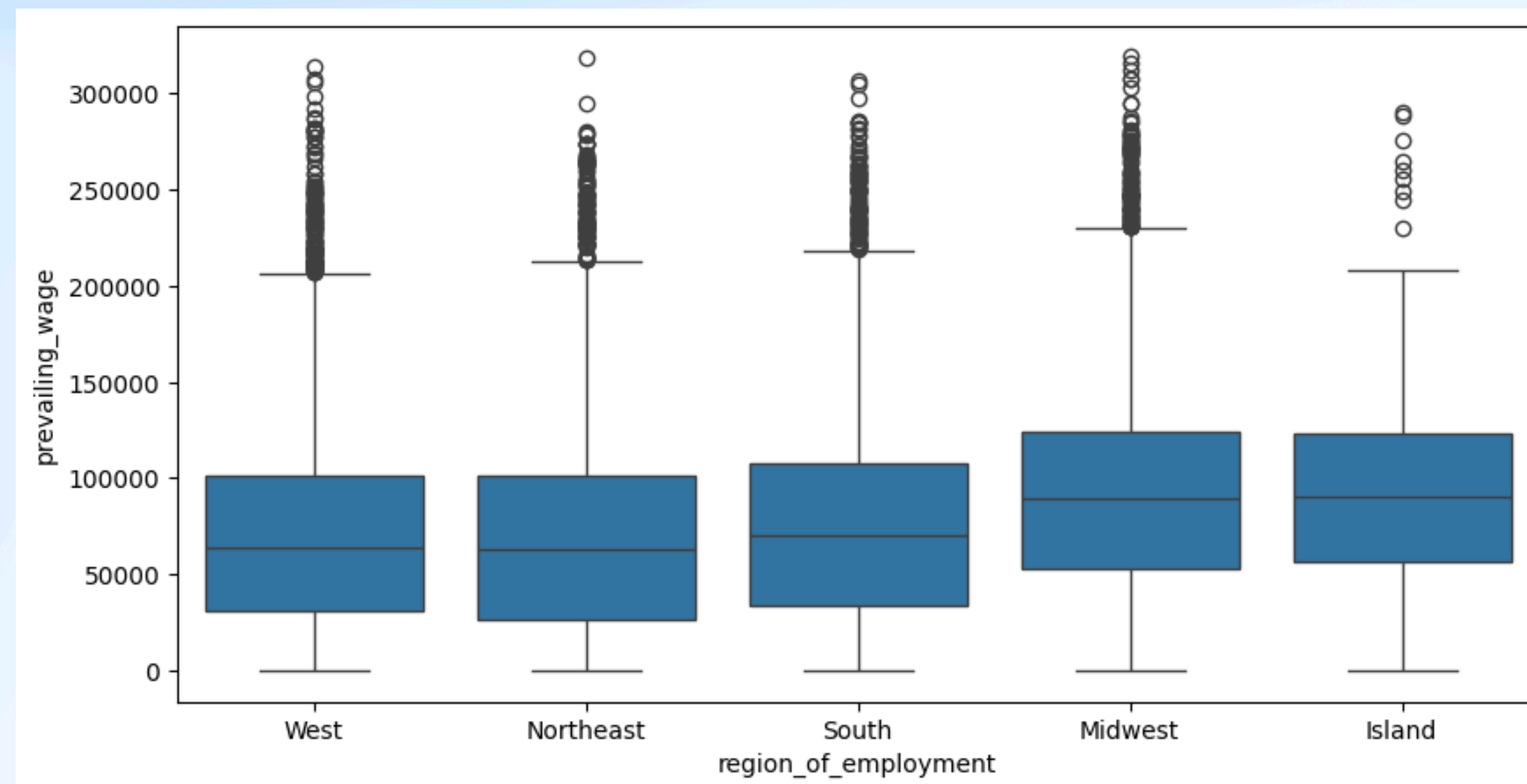
Job experience does play a factor in approval rates. We see that employees with job experience get accepted at ~75% compared to employees with no job experience at ~57%.



EDA Results - Bivariate Analysis

Prevailing Wage & Region of Employment

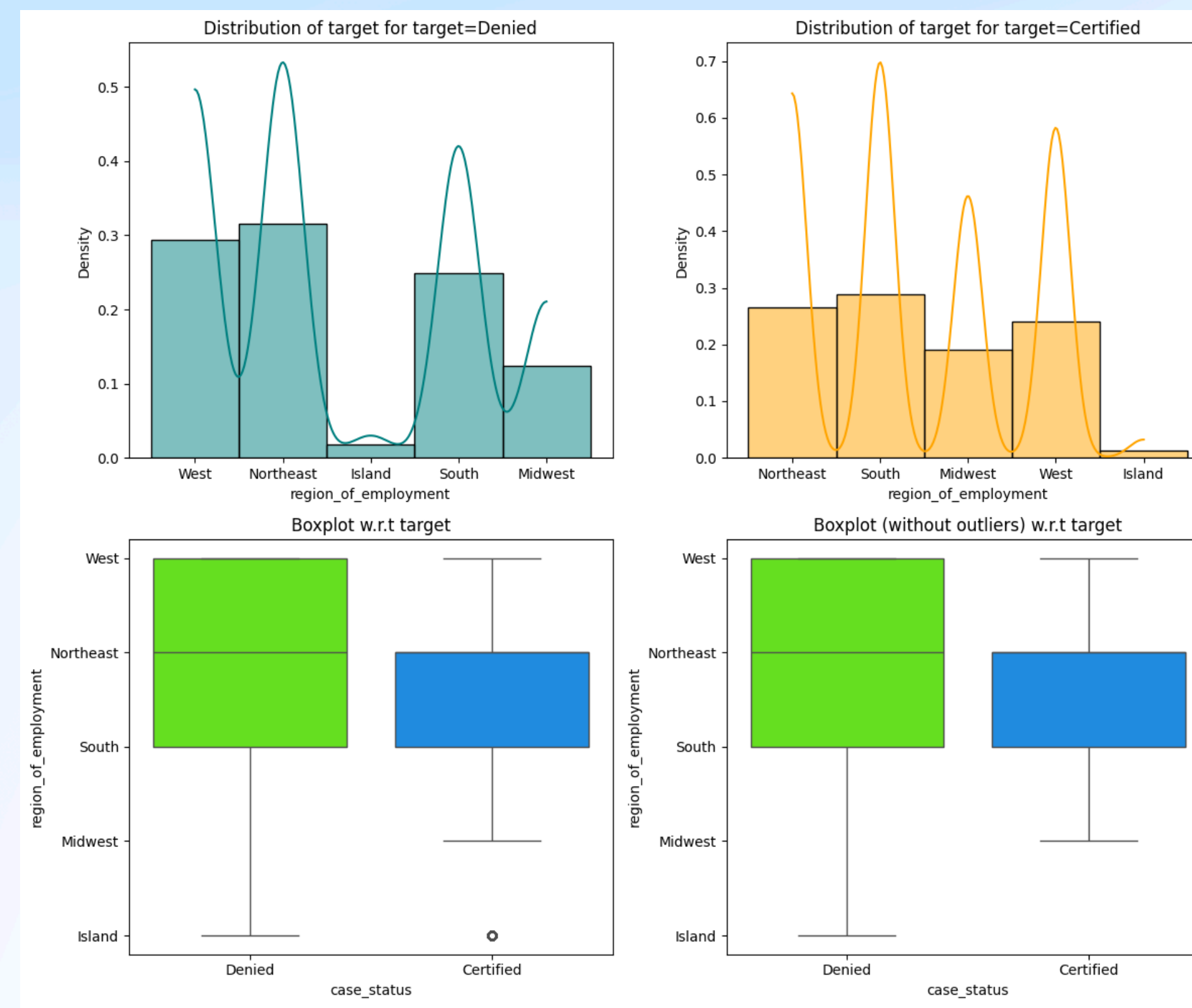
We see that the Island and Midwest region lead the country in median prevailing wages with both being right under \$100,000. West and Northeast regions have the lowest median wages being around \$60,000.



EDA Results - Bivariate Analysis

Prevailing Wage & Region of Employment

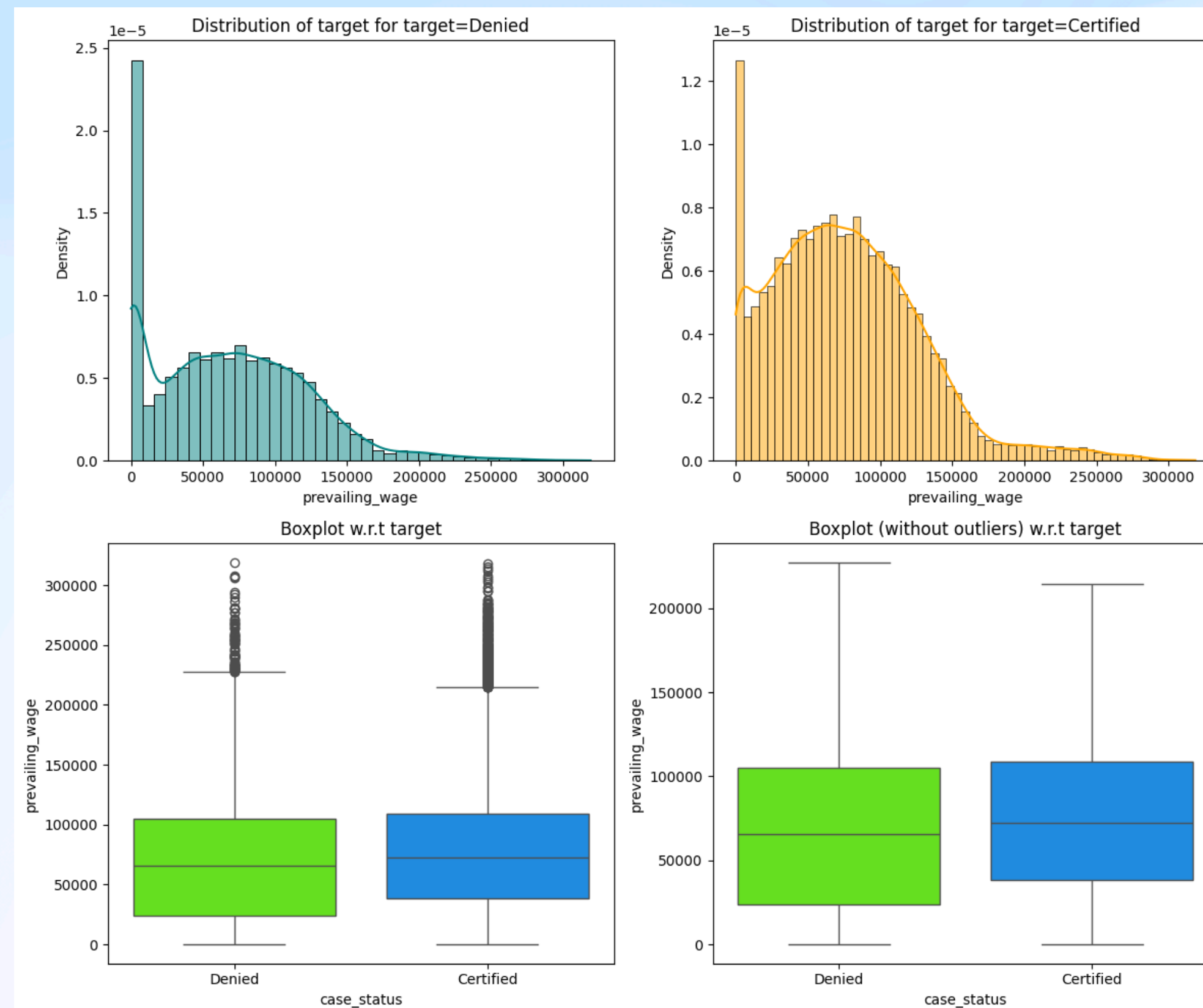
Looking at the denied cases we see the Island region has the least denials and Northwest region has the highest number of denials. South and Northeast regions have the highest number of approvals and island has the lowest number of approvals.



EDA Results - Bivariate Analysis

Prevailing Wage & Visa Approval

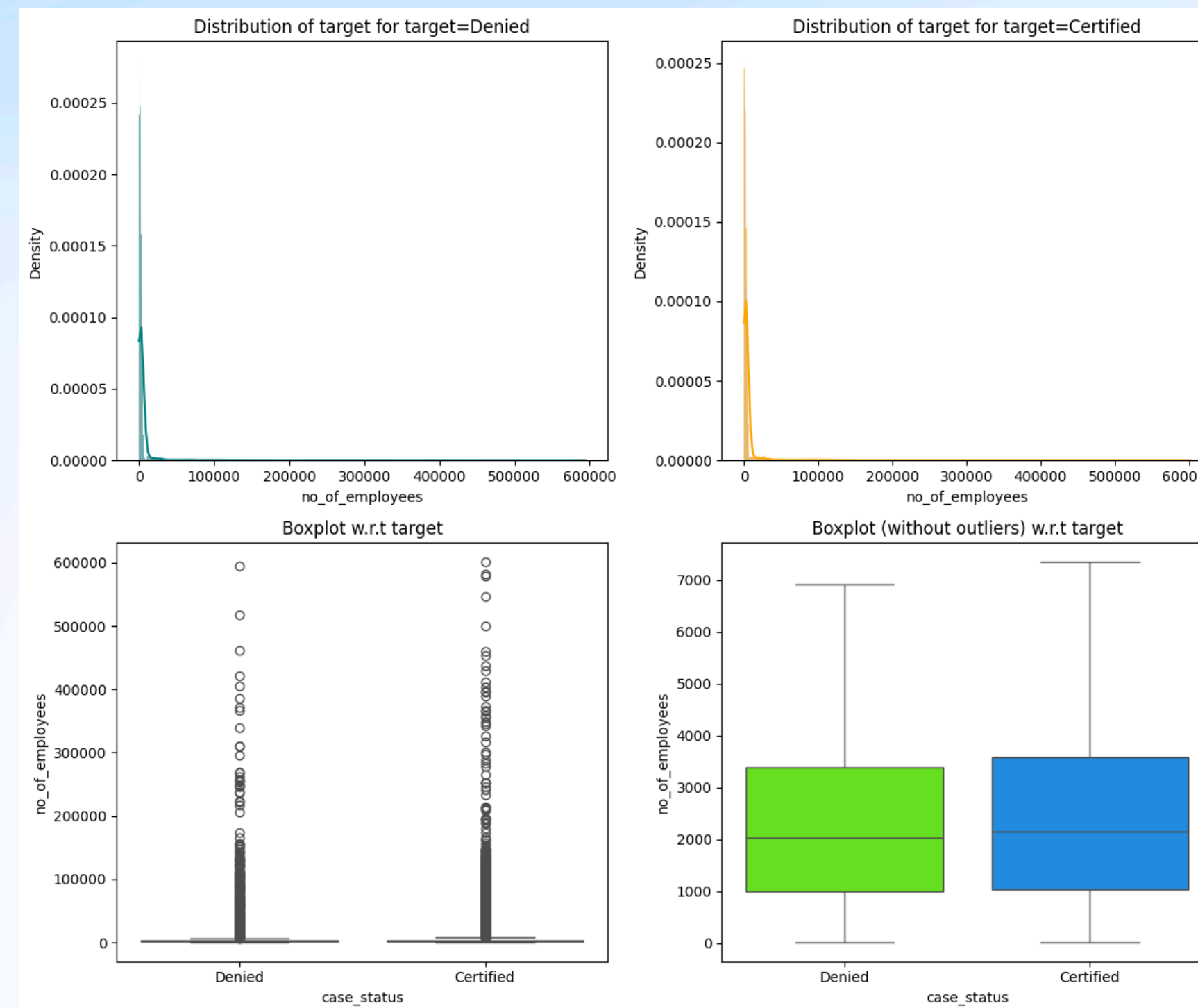
We can see a significant number of denied cases are associated with low prevailing wages. Looking at the approved cases we see there are a lot of accepted cases between \$50,000 - \$100,000.



EDA Results - Bivariate Analysis

Number of Employees & Visa Approval

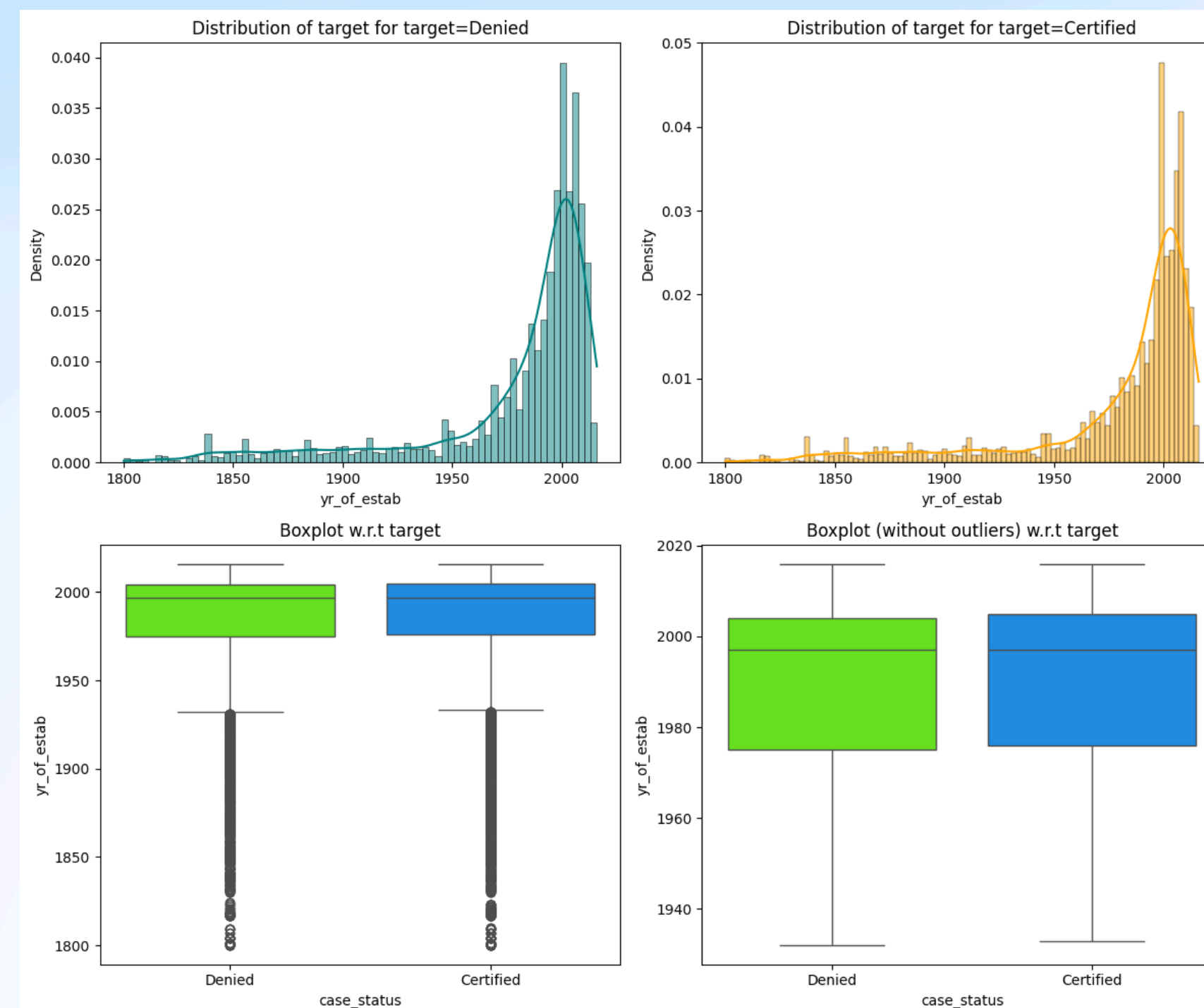
The median number of employees for companies who have accepted visa applications is about 2,100 employees compared to about 2,000 for companies who have denied visa applications. Larger companies do have a higher approval rate even though its not by a lot.



EDA Results - Bivariate Analysis

Year of Establishment & Visa Approval

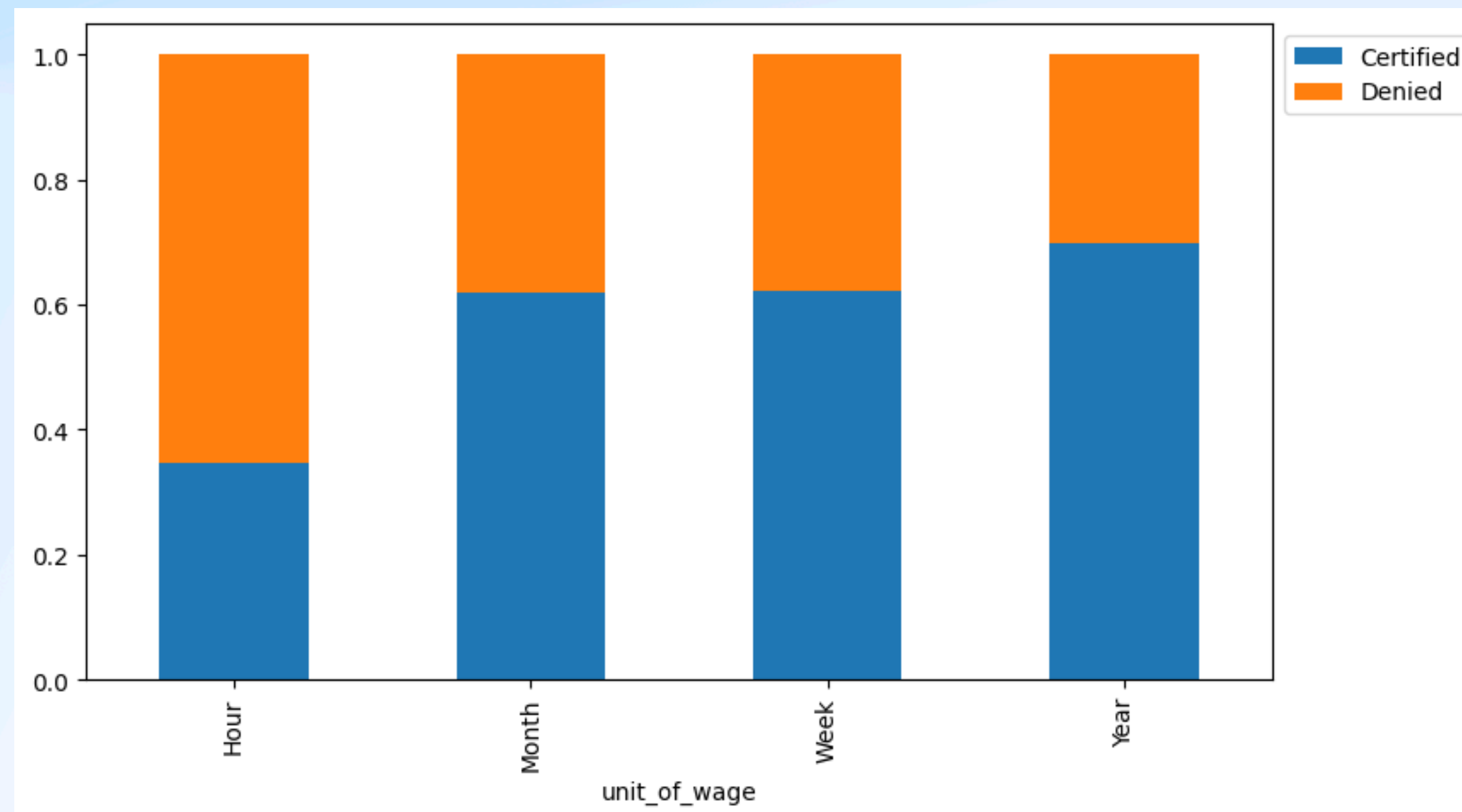
There is no correlation between 'year of establishment' and visa approval. Both tend to have a high approval and denial rate for companies that were established after 1950.



EDA Results - Bivariate Analysis

Unit of Wage & Visa Approval

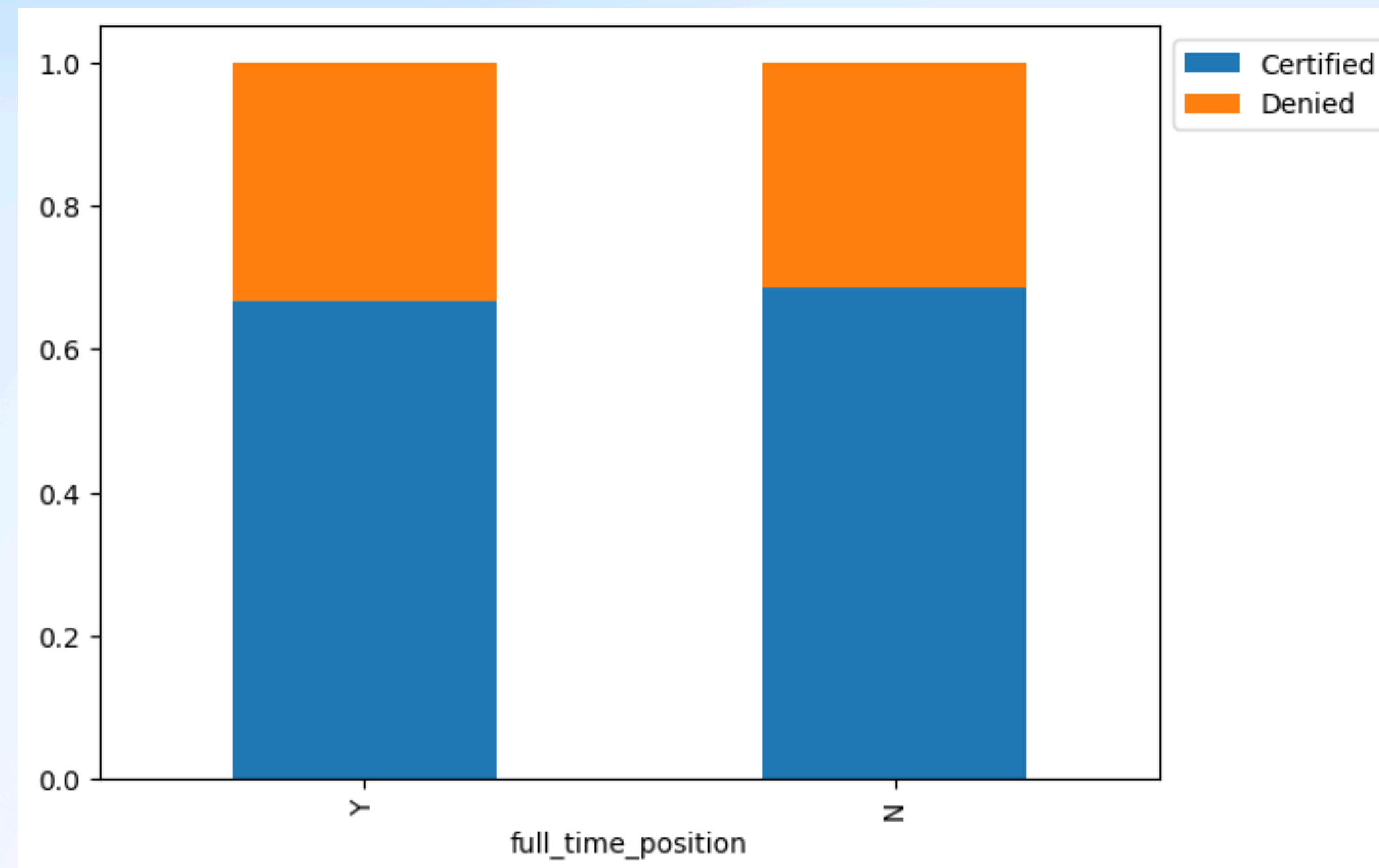
We can see the higher unit of wage the more likely the applicant will get his Visa accepted. Yearly unit of wage leads all units with ~70% approval and Hour unit of wage has the least number of approvals with ~35%.



EDA Results - Bivariate Analysis

Full Time Position & Visa Approval

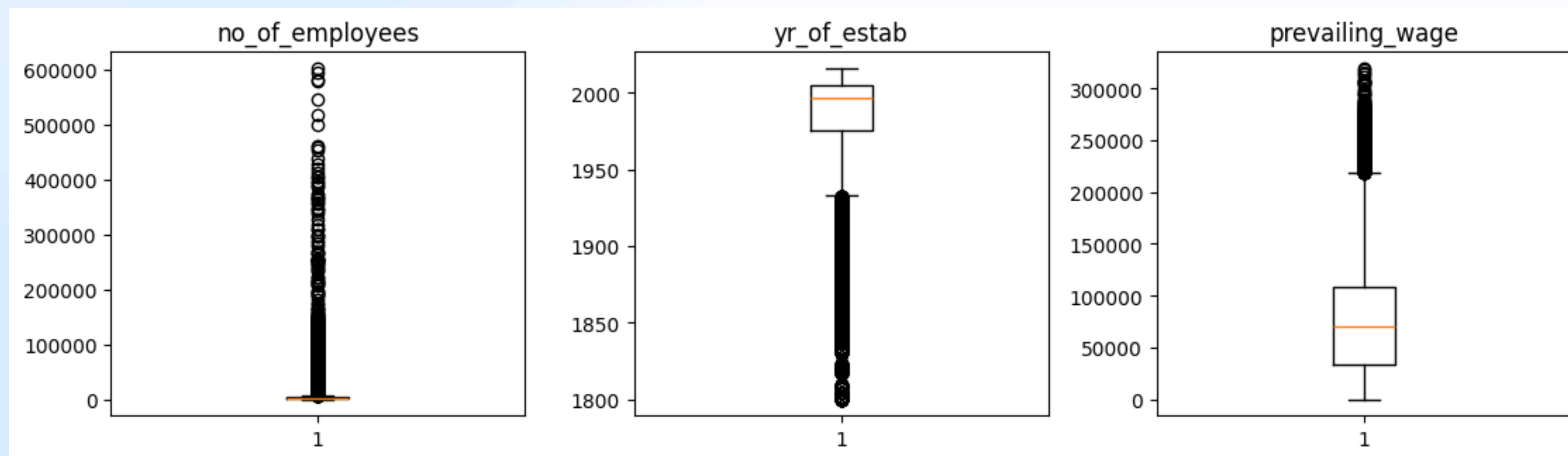
There is no correlation between 'full time position' and visa approvals. Both full and part time have ~70% approval odds.



Data Pre-processing

Outlier Check

Looking at 'number of employees' we see there is a large amount of outliers going all the way to 600,000. For 'year of establishment' majority of the companies were founded after the 1930s. There are some companies founded before going all the way down to 1800. We see the majority in 'Prevailing Wage' is between 0 - ~\$225,000.



Data Preparation for Modeling

We need to prepare the data for modeling. We can do this by splitting the data into train, test and validation sets.

- The shape of the 3 sets are:
 - Training set: (15288,21)
 - Validation set: (5096,21)
 - Test set: (5096, 21)
- Percentage of classes in training set are:
 - Certified: 0.667844
 - Denied: 0.332156
- Percentage of classes in validation and test set are both:
 - Certified 0.667975
 - Denied: 0.332025

Model Building

Original Data

Looking at 5 models below using the original data we can observe that the Random Forest and Bagging have memorized the rows in the training set including the noise and outliers. This is why you see a big difference on the validation set. Gradient boosting will be the best model here because it has such a small difference and the highest Val F1 score.

Original Data

Model	Train F1	Val F1	Difference
Bagging	0.9892	0.7737	0.2155
Random Forest	1.0000	0.8050	0.1950
AdaBoost	0.8204	0.8180	0.0024
Gradient Boosting	0.8291	0.8266	0.0025
XGBoost	0.8963	0.8079	0.0884

Model Building

Oversampled Data

Looking at 5 models below using the oversampled data we can observe that the Random Forest and Bagging models are overfitted because of the huge difference between Train F1 and Val F1. AdaBoost and Gradient Boosting models are robust because the Val F1 scores outperformed the Train F1 scores.

Oversampled Data

Model	Train F1	Val F1	Difference
Bagging	0.9875	0.7665	0.2210
Random Forest	1.0000	0.7965	0.2034
AdaBoost	0.8005	0.8195	-0.0190
Gradient Boosting	0.8072	0.8173	-0.0101
XGBoost	0.8709	0.8129	0.0579

Model Building

Undersampled Data

Looking at 5 models below using the Undersampled data we can observe that the Random Forest and Bagging models are severely overfitted because of the huge difference between Train F1 and Val F1. More than the oversampled data. AdaBoost and Gradient Boosting models are robust because the Val F1 scores out performed the Train F1 scores.

Undersampled Data

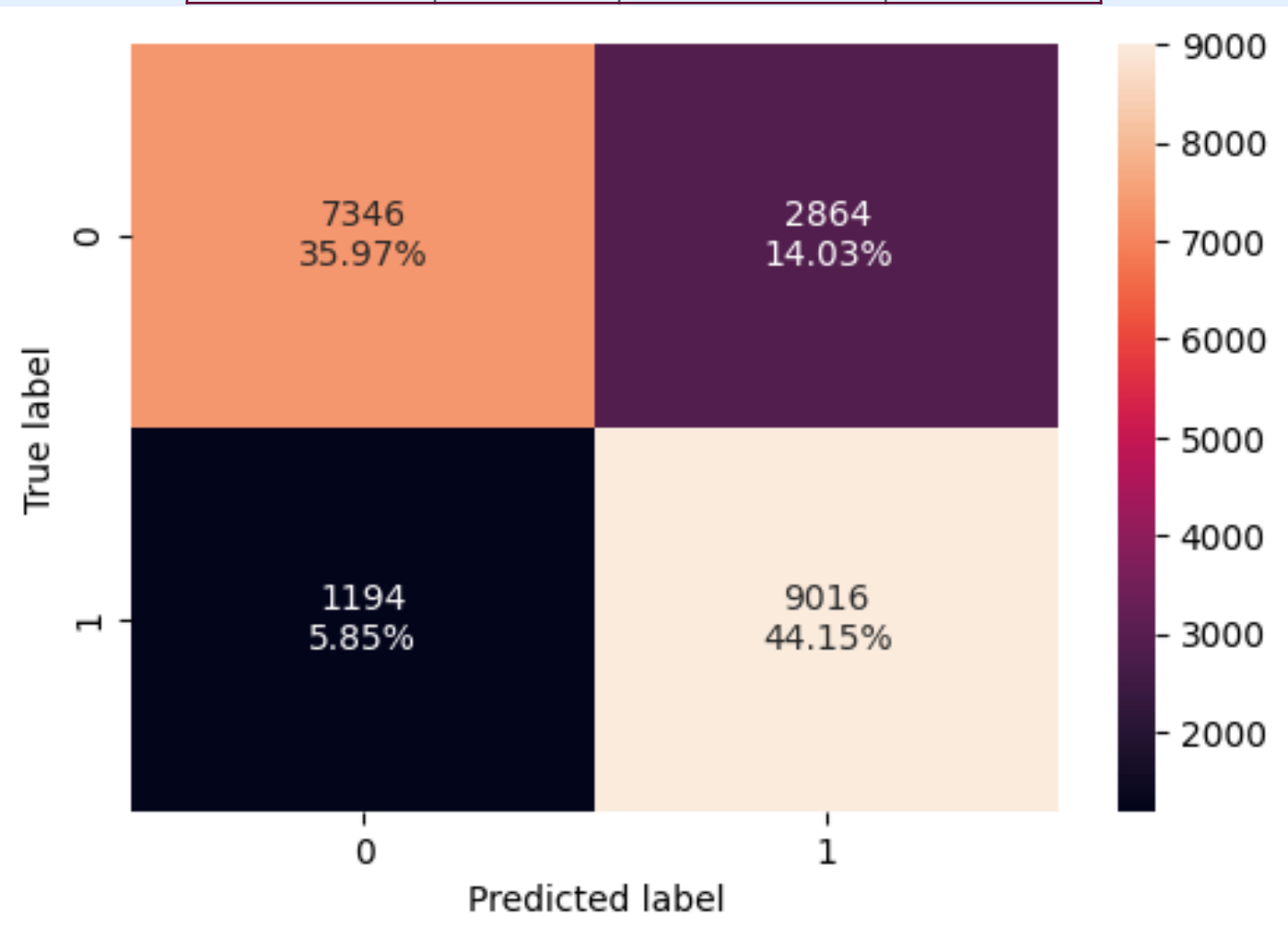
Model	Train F1	Val F1	Difference
Bagging	0.9804	0.7057	0.2747
Random Forest	1.0000	0.7417	0.2583
AdaBoost	0.7015	0.7660	-0.0645
Gradient Boosting	0.7281	0.7766	-0.0485
XGBoost	0.8720	0.7459	0.1261

Hyperparameter tuning

Random Forest

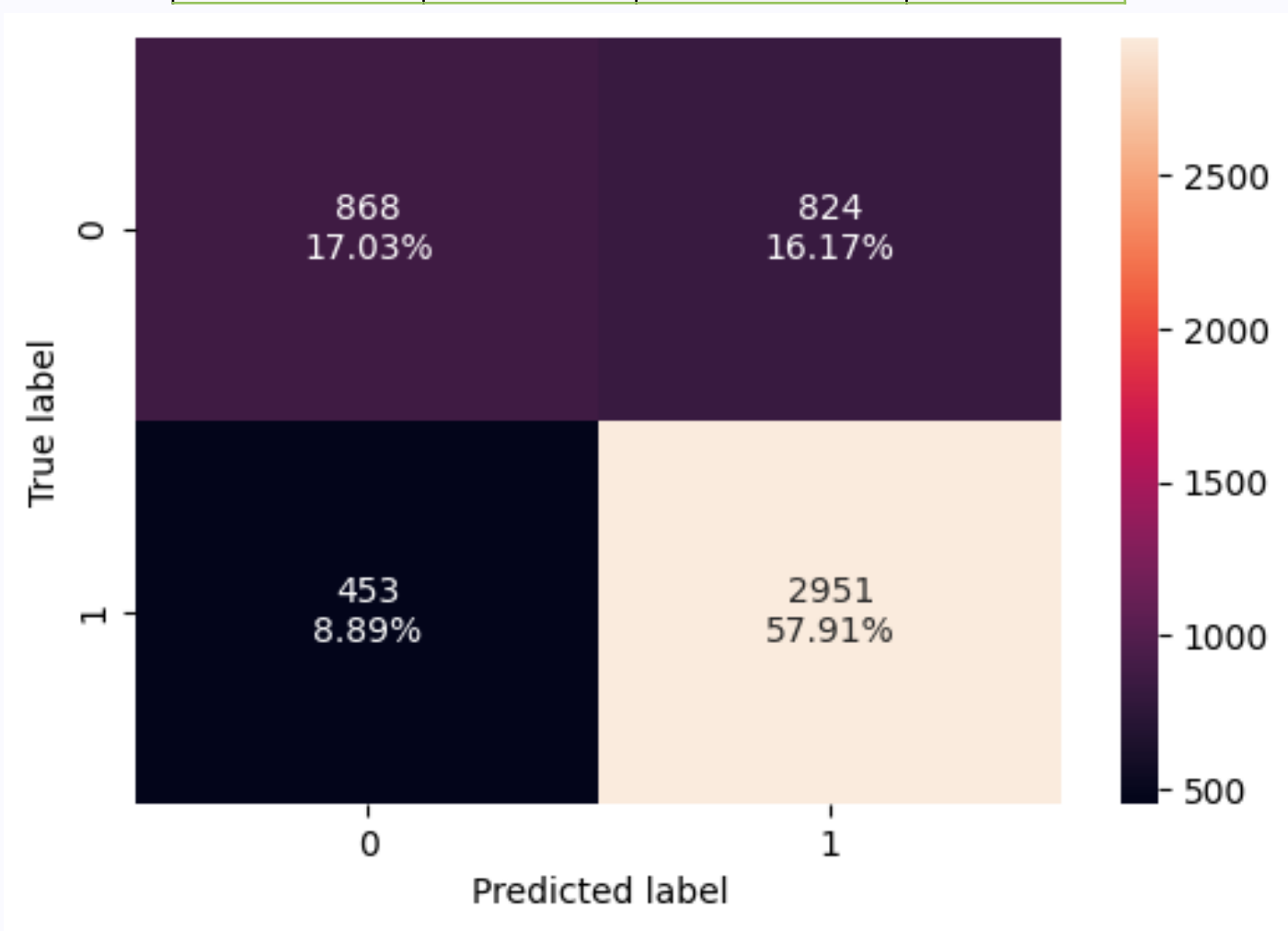
Below is the training set using Random Forest. We see that the model predicted 2,864 cases as certified but these were actually denied. This model is better at identifying certified cases than denied ones. We know this because the recall is high.

Accuracy	Recall	Precision	F1
0.801273	0.883056	0.758923	0.816297



Below is the validation set using Random Forest. We can see the model is good at finding certified cases because the recall is very good. This model incorrectly labeled 824 denied cases as certified. Overall the model is better at predicting certified cases than denied cases.

Accuracy	Recall	Precision	F1
0.749411	0.866921	0.781722	0.82212

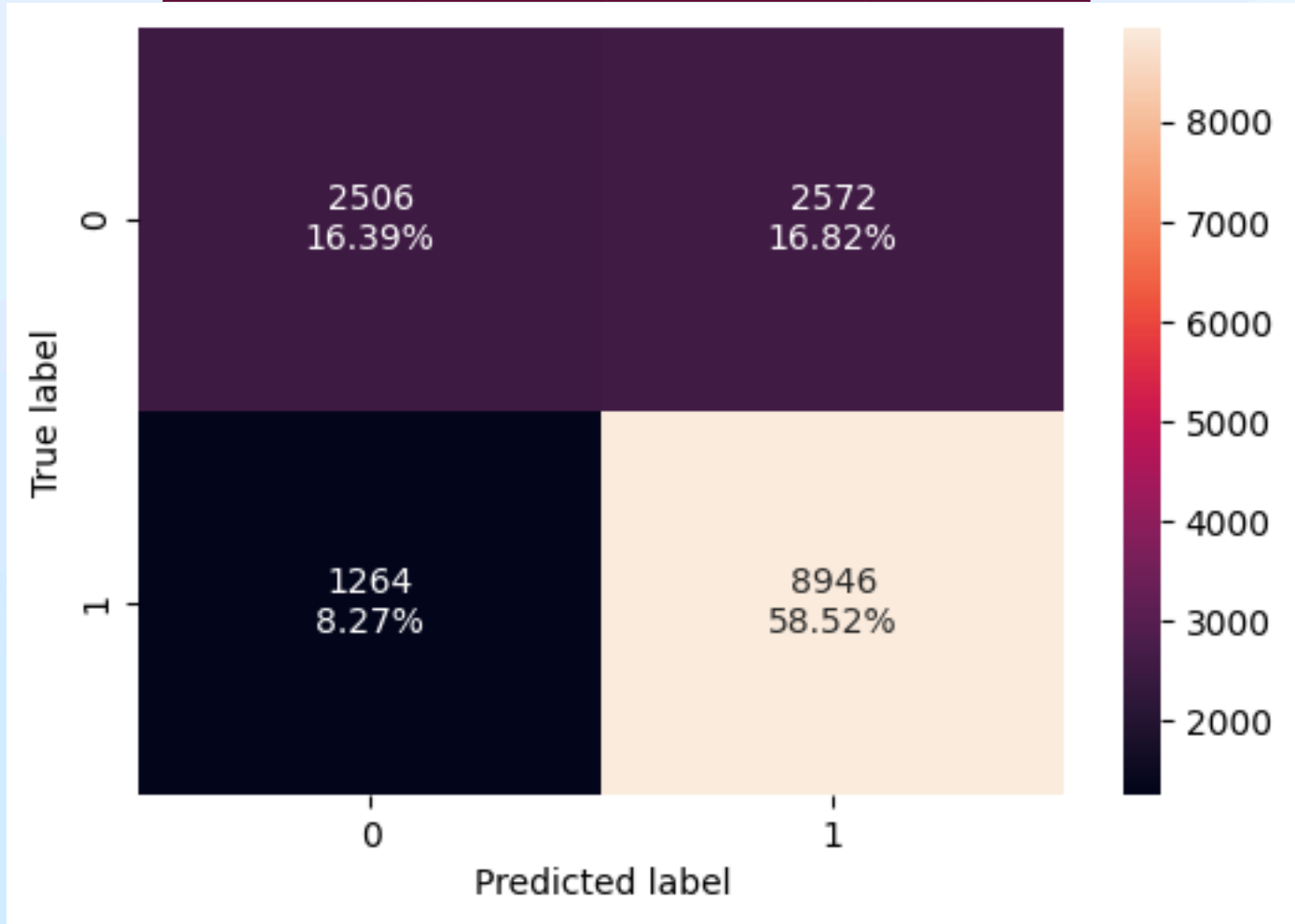


Hyperparameter tuning

AdaBoost

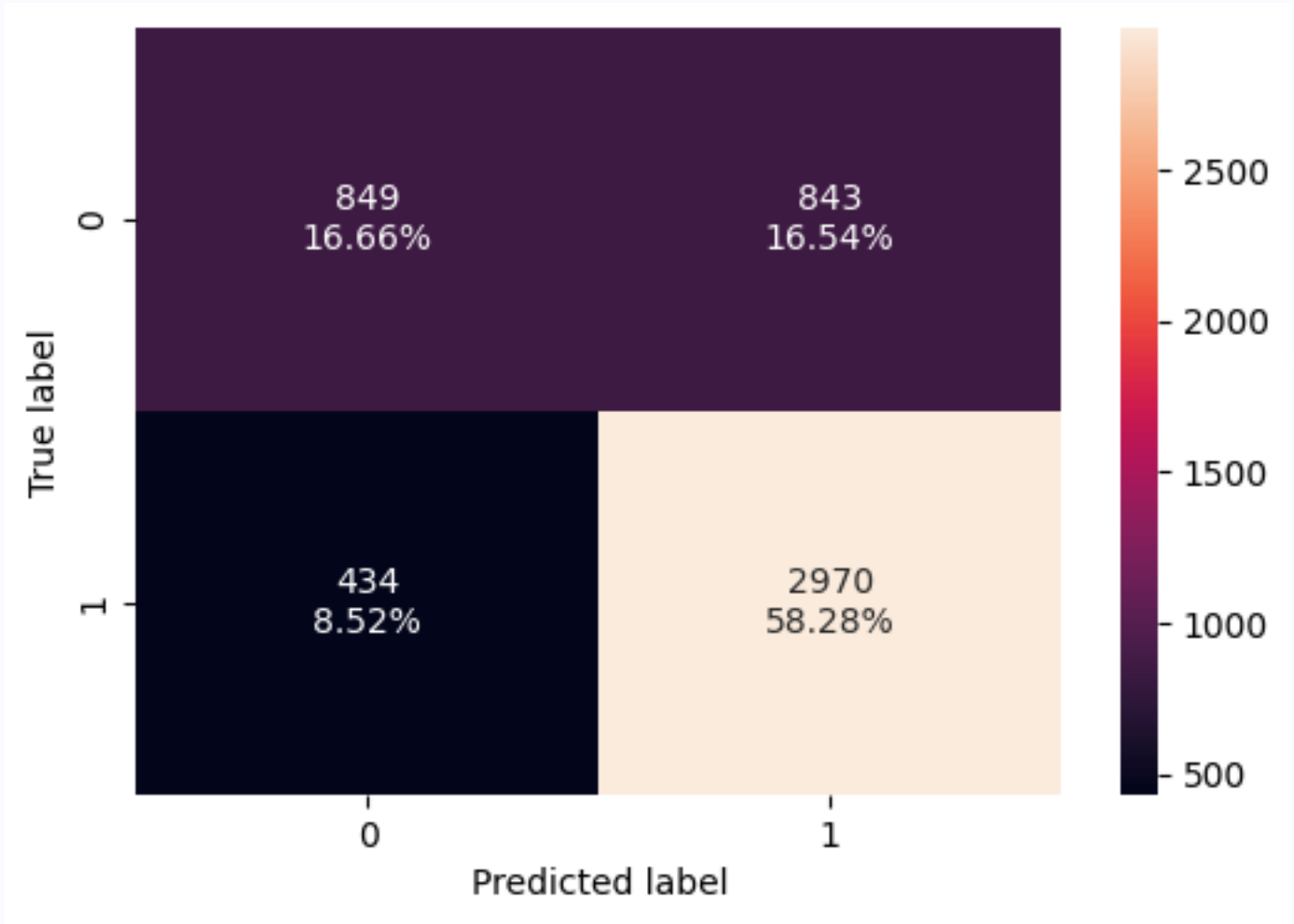
Below is the training set using AdaBoost. We see that the model predicted 2,572 cases as certified but these were actually denied. The model is predicting 2,506 true negatives. The model is giving us more false positives than true negatives. However, the model is very successful at identifying certified cases.

Accuracy	Recall	Precision	F1
0.749084	0.8762	0.776697	0.823454



Below is the validation set using AdaBoost. We can see the model is good at finding certified cases and that number is 2,970. This model incorrectly labeled 843 denied cases as certified. The model is great at certifying cases because it has a high recall of ~87.2%

Accuracy	Recall	Precision	F1
0.749411	0.872503	0.778914	0.823057

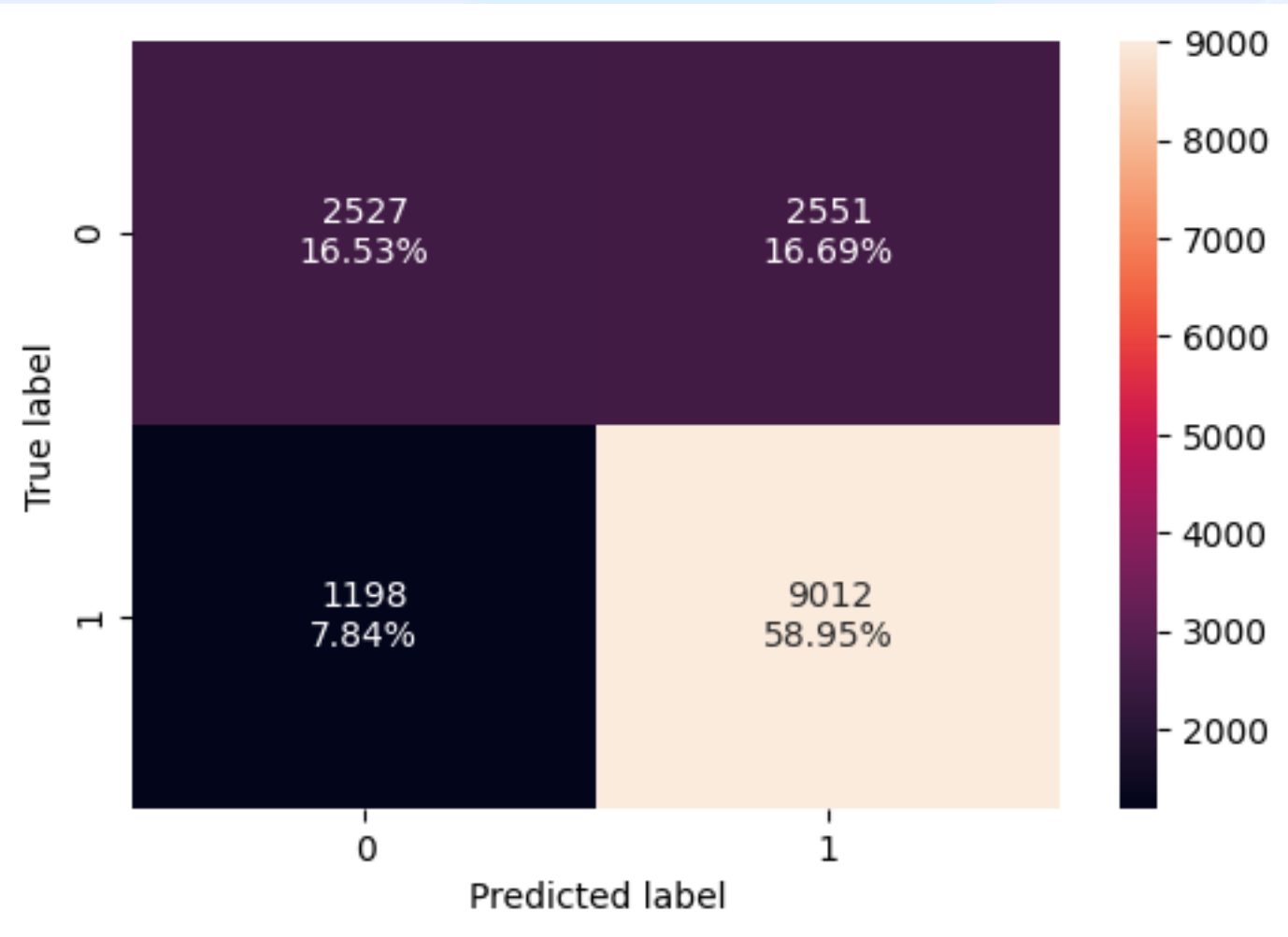


Hyperparameter tuning

Gradient Boosting

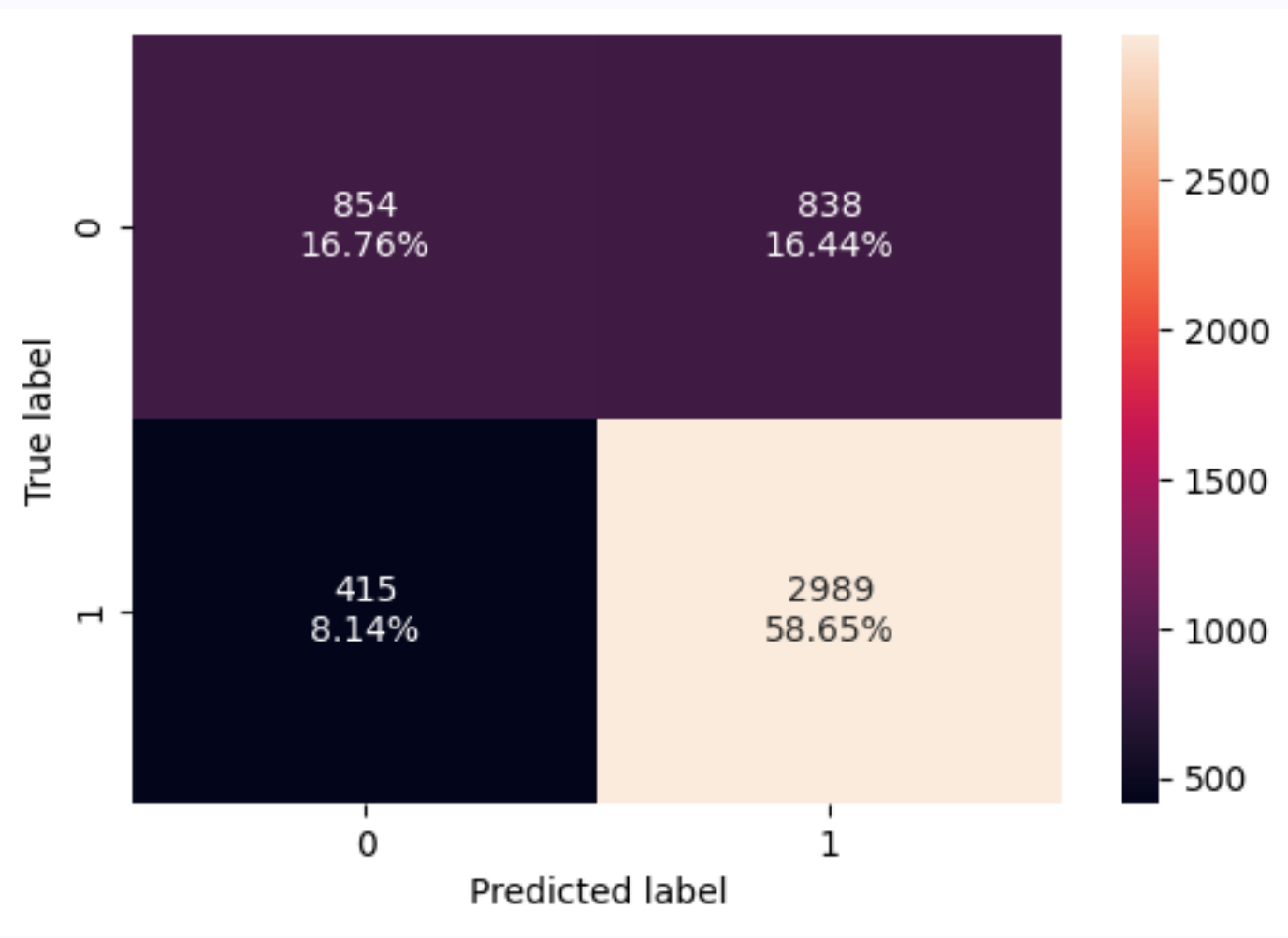
Below is the training set using Gradient Boosting. We can observe that 9,012 cases were correctly identified as certified. The model was successful in identifying about half of the denials which was 2,527. Having a recall of ~88.3% tells us the model finds almost all approvals.

Accuracy	Recall	Precision	F1
0.754775	0.882664	0.779383	0.827814



Below is the validation set using Gradient Boosting. We can see the model is good at finding certified cases and that number is 2,989. This model incorrectly labeled 838 denied cases as certified. Overall the models recall is very good at ~87.8%

Accuracy	Recall	Precision	F1
0.754121	0.878085	0.78103	0.826718

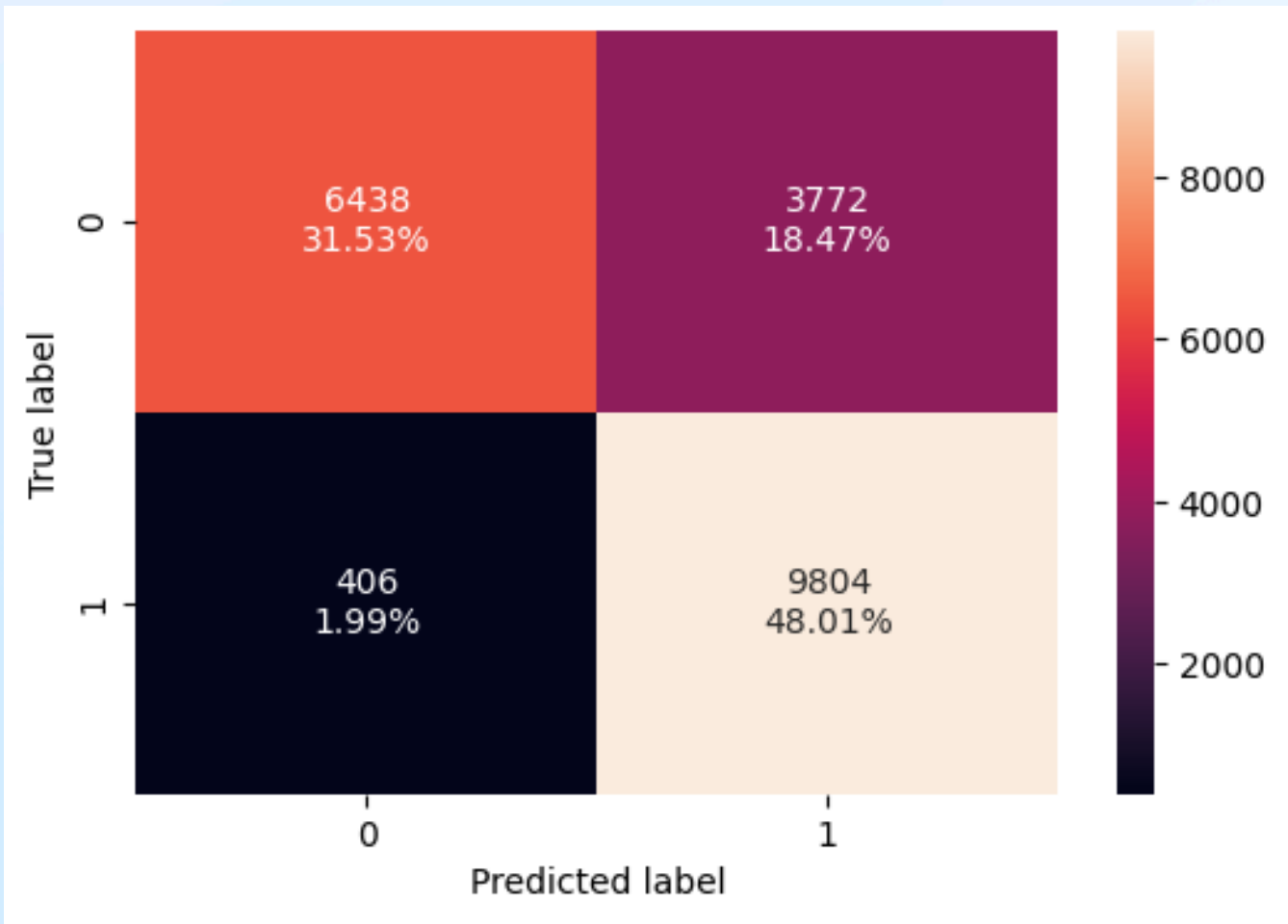


Hyperparameter tuning

XGBoost

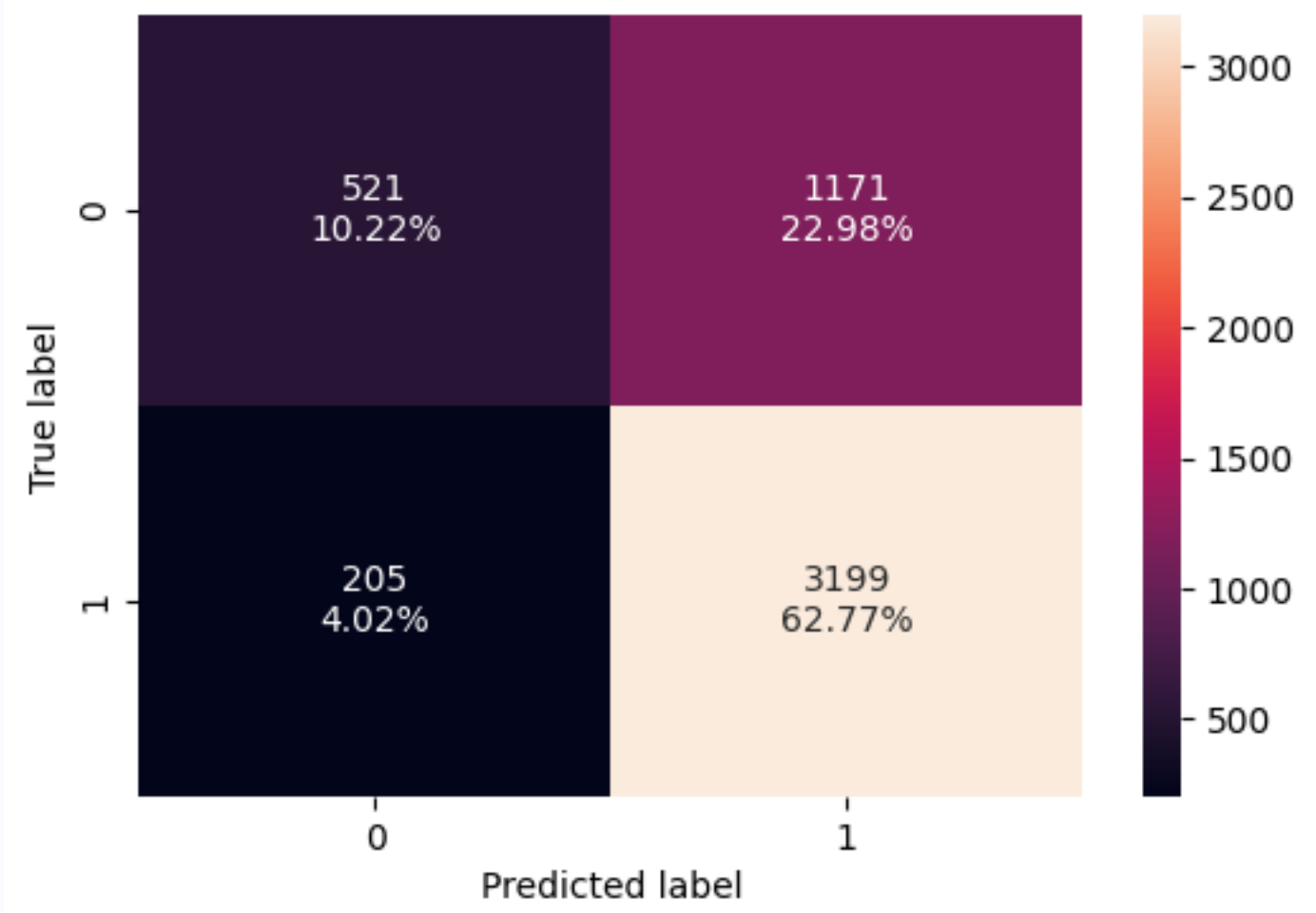
Below is the training set using XGBoost. We can observe that the recall is 96%. This means the model is almost perfect when it comes to finding every certified case in the data. We also see 3722 cases were predicted as certified even though they were denied. This can mean that the model is very optimistic and approves more cases than it should.

Accuracy	Recall	Precision	F1
0.795397	0.960235	0.722157	0.82435



Below is the validation set using XGBoost. We can see the model is good at predicting certified cases. It does however have a high error rate and you can see that with the 1,171 cases it predicted as certified and were actually denied.

Accuracy	Recall	Precision	F1
0.729984	0.939777	0.732037	0.823



Model Comparison & Final Model Selection

Training Set

Below is a chart where I compare all 4 training sets. We can observe that Tuned Gradient Boost Classifier has the highest F1 score. We see XGBoost has a very high recall and gets most all certified cases correctly. Tuned Random Forest has the highest accuracy out of all 4 models. It's F1 score is the lowest. This may be because this model is likely better at predicting denied cases compared to certified cases. Tuned Adaboost Classifier does have a high F1 score and high precision. This makes the model better than Random Forest. We will choose XGBoost Classifier.

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.801273	0.749084	0.754775	0.795397
Recall	0.883056	0.876200	0.882664	0.960235
Precision	0.758923	0.776697	0.779383	0.722157
F1	0.816297	0.823454	0.827814	0.824350

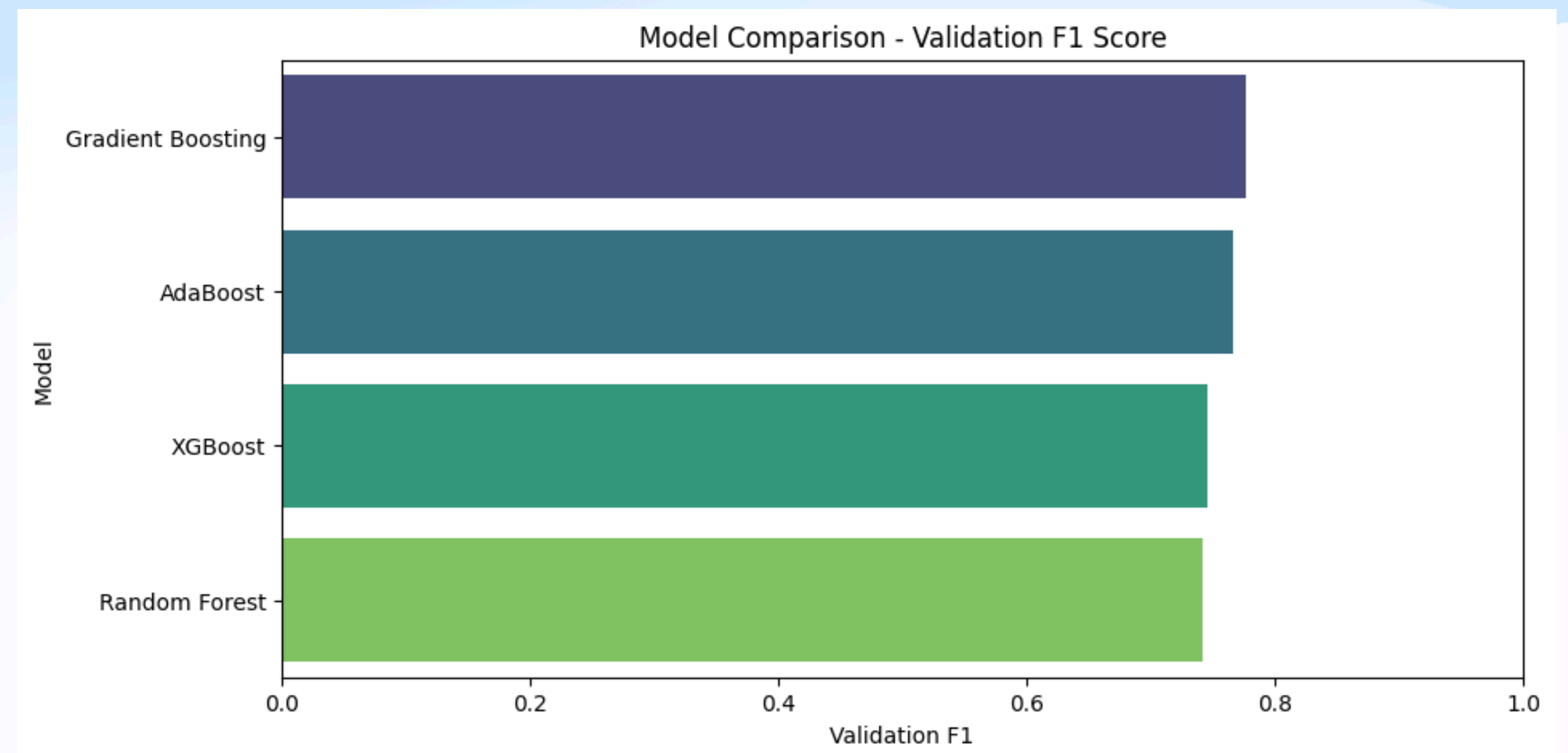
Model Comparison & Final Model Selection

Validation Set

Below is a graph where all 4 Validation F1 Scores are listed from highest to lowest. We see Gradient Boosting has the highest F1 score out of all the models. However, XGBoost does have a high F1 score and is better at predicting certified visas. XGBoost will be the final model we choose.

Validation Set Performance Comparison:

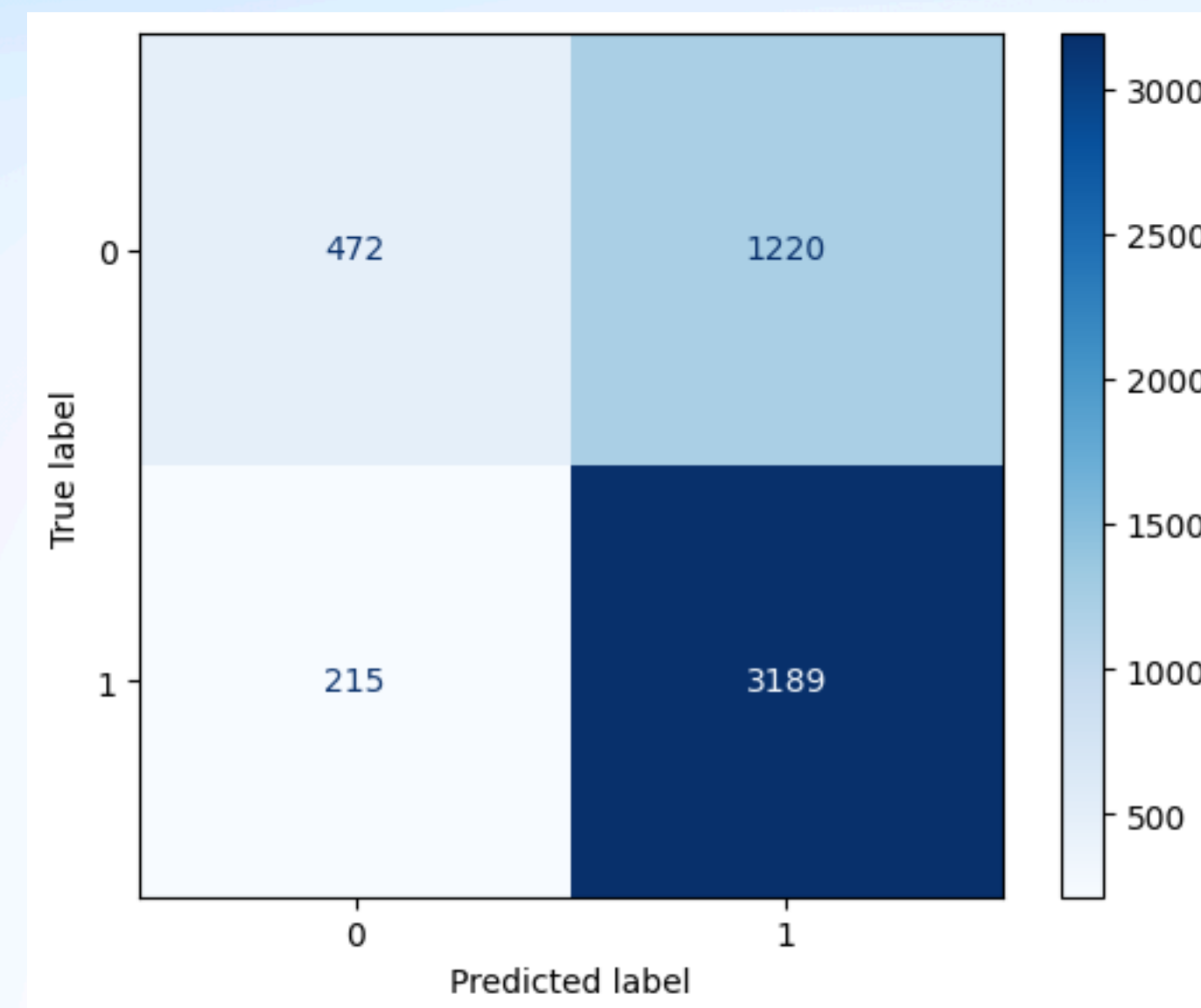
Model	Validation F1
Gradient Boosting	0.776596
AdaBoost	0.765991
XGBoost	0.745930
Random Forest	0.741722



Performance of Best Model on Test Data

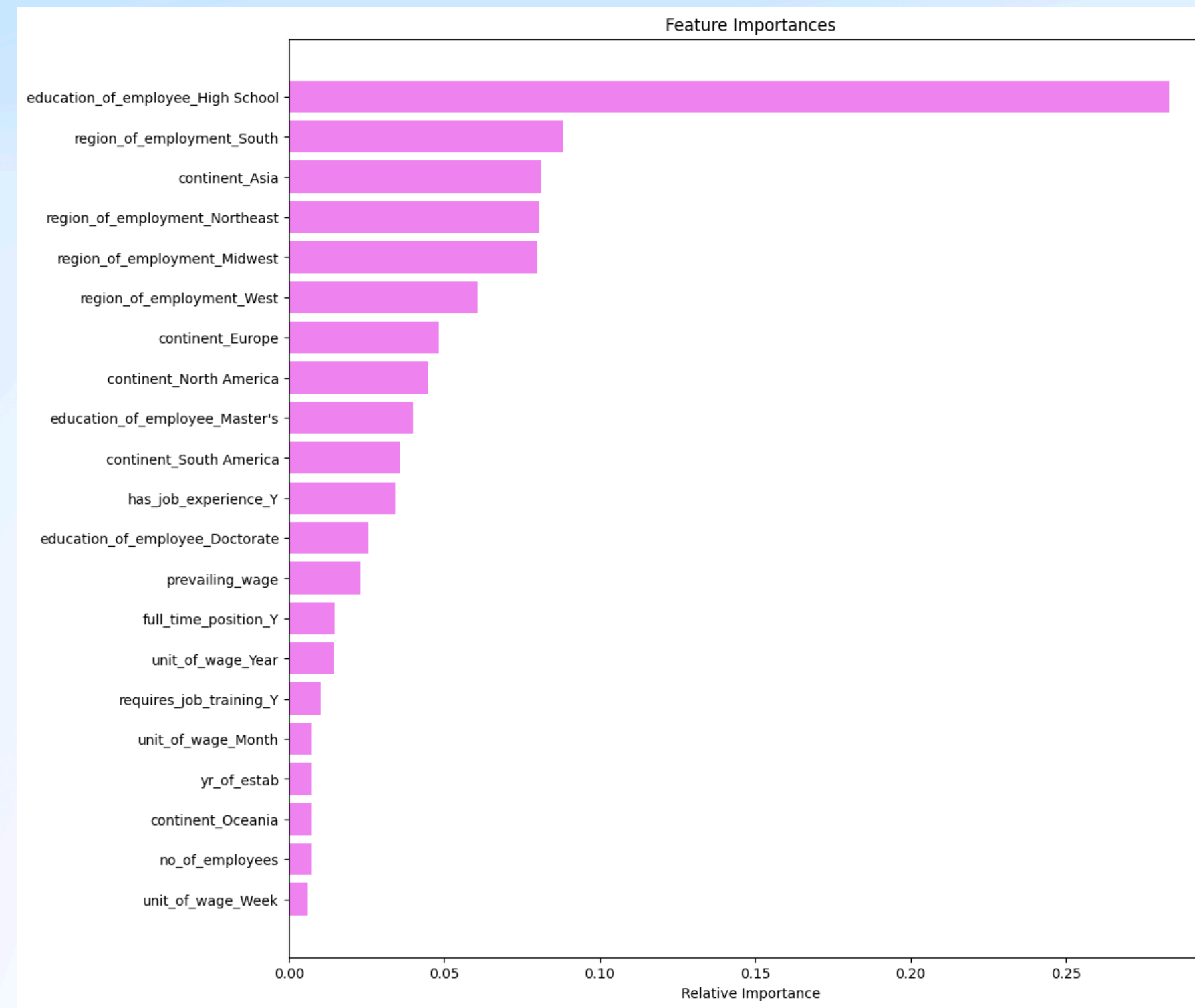
XGBoost Classifier

Below is a graph where we see the model predicted 3189 cases correctly as certified. It missed only 215 certified cases that should have been approved giving it a 97% Recall. This model also has a ~81% F1 Score and ~72% Precision. The XGBoost Classifier is the best model.



Important Features of the Final Model

Below we can see the most important features for our model are 'education of employee High School' and 'region of employment South' and 'continent Asia'. We can make an observation that the model is using high school education as a way to say no automatically and move past those applicants. We can also observe that the south region in the USA approves a lot of applicants and most of the approved applicants who are coming to that region are from Asia.



THANK YOU!